TAMR-DTI: TARGET-AWARE MODULATION AND MAMBA-BASED REPRESENTATION REFINEMENT FOR DRUG-TARGET INTERACTION PREDICTION

Wei Li[1], **Wei Liu**[2*], **Xiaoqing Hu**[3*], **and Shenghui Shi**[1*]
[1] Beijing University of Chemical Technology, Beijing, China
[2] Academy of Military Medical Sciences, Beijing, China
[3] Peking University Third Hospital, Beijing, China
[*]Corresponding authors: liuweilcf@163.com; huxiaoqingbd01@sina.com; shish@mail.buct.edu.cn

ABSTRACT

Recent multimodal drug-target interaction (DTI) methods have improved prediction by integrating pretrained sequence representations, molecular topology, and three-dimensional ligand geometry. However, many pipelines still summarize candidate conformers as target-independent drug features and defer cross-modal conditioning to late fusion, so the same ligand geometry can be used for proteins with different binding contexts. In addition, common long-sequence modeling strategies are often introduced as generic representation refinement modules; when used without preserving explicit drug-target matching, they may weaken the pairwise interaction bias required by DTI prediction. To address these limitations, we propose TAMR-DTI, a target-aware multimodal framework that improves a strong BiIntention-based backbone through two complementary innovations. First, a target-aware modulation mechanism uses protein context to reweight energy-ranked drug conformers and uses fused ligand cues to condition protein token encoding through FiLM, yielding target-conditioned ligand geometry and ligand-aware protein representations before final interaction modeling. Second, a gated bidirectional Mamba residual block is applied only to the ordered protein token stream before BiIntention, strengthening long-range protein sequence refinement while leaving explicit BiIntention matching to model drug-target alignment. We evaluate TAMR-DTI on BindingDB, BioSNAP, Human, and C. elegans. Using repeated-seed benchmark experiments, module ablations, conformer-count comparisons, and conformer-weight diagnostics, we examine both predictive performance and mechanism-level contributions. The results indicate improved AUROC and AUPRC across all four benchmarks, with the clearest gain on BioSNAP, while Human remains a higher-variance setting for fixed-threshold prediction. Overall, TAMR-DTI provides a practical way to combine protein-conditioned molecular geometry with protein-only state-space refinement, offering a more interaction-aware design for robust DTI prediction.

1 INTRODUCTION

Drug-target interaction (DTI) prediction is a fundamental problem in computational drug discovery, because it connects candidate molecules with potential therapeutic targets before expensive biochemical validation. Reliable DTI models can accelerate chemical-genomics screening, target identification, and drug repositioning by narrowing the search space of plausible compound-protein pairs (Keiser et al., 2007; Yamanishi et al., 2008; Liu et al., 2007). The task is intrinsically multimodal: interaction signals depend jointly on molecular chemistry, three-dimensional ligand geometry, protein sequence context, and the pair-specific compatibility of the two modalities. An effective model therefore must encode each side with appropriate inductive biases while preserving the interaction structure that distinguishes binding from non-binding pairs.

Modern DTI models have moved from hand-crafted descriptors and similarity inference toward learned multimodal representations. Pretrained molecular and protein language models, graph neural networks, geometric encoders, and attention-based interaction modules now make it possible to combine sequence semantics, molecular topology, ligand conformer geometry, and explicit cross-modal matching in one prediction pipeline (Chithrananda et al., 2020; Elnaggar et al., 2022; Satorras et al., 2021; Peng et al., 2025; Ji et al., 2026). This progress has strengthened the representational capacity of DTI systems, but it also raises a sharper modeling question: how should pair-specific conditioning be introduced before the final interaction decision?

Two design gaps remain only partially answered within this trajectory. The first concerns how three-dimensional ligand geometry should enter a multimodal pipeline. Most existing systems treat candidate conformers as a target-agnostic property of the drug, although whether a particular conformer is informative typically depends on the protein context with which the ligand is paired (Satorras et al., 2021; Ji et al., 2026). If conformer information is collapsed before seeing the target, the geometry branch may dilute conformations that are useful for one protein with conformations that are irrelevant for another. Addressing this issue would turn ligand geometry from a static drug descriptor into pair-level evidence for DTI prediction. TAMR-DTI targets this gap by using protein context and conformer energy to score valid conformers, aggregate EGNN-encoded conformer tokens, and form a target-conditioned drug representation that later modulates protein encoding.

The second gap concerns where long-range sequence modeling should sit relative to cross-modal matching. Structured state-space models offer linear-time long-sequence dynamics that are competitive with self- attention (Gu et al., 2022; Gu & Dao, 2023), and a recent hybrid Mamba-attention design has begun to explore this direction in DTI (Shui et al., 2025). However, applying sequence refinement to mixed drug-protein token streams can impose artificial sequential structure on heterogeneous drug tokens, while replacing explicit cross-modal attention may remove the pairwise matching bias that distinguishes DTI from generic representation learning (Bai et al., 2023; Peng et al., 2025). Addressing this issue would let the model benefit from efficient protein sequence refinement without weakening the drug-target alignment module. TAMR-DTI therefore applies a gated bidirectional Mamba residual block only to the ordered protein token stream and then passes the refined protein representation to BiIntention for explicit cross-modal matching.

Motivated by these gaps, a DTI model should satisfy three requirements. First, ligand geometry should be selected or aggregated under the context of the paired target, rather than summarized once for the ligand alone. Second, protein representations should be allowed to depend on the ligand, because the same sequence can expose different interaction evidence for different small molecules. Third, long-range protein refinement should complement, rather than replace, an explicit cross-modal matching module. These requirements motivate a design that separates target-aware representation construction from final drug-target alignment.

To this end, we propose TAMR-DTI, a target-aware multimodal framework built on a BiIntention-based cross-modal matching backbone (Peng et al., 2025). TAMR-DTI uses protein context to reweight candidate ligand conformers, uses the fused drug representation to modulate protein token encoding, and applies a gated Mamba residual block only to the ordered protein token stream before BiIntention (Gu & Dao, 2023). In this way, the model introduces pair-specific conditioning before final matching while keeping state-space sequence refinement and drug-target alignment as distinct components.

The main methodological contributions of this work are summarized as follows:

- We formulate a target-aware modulation strategy that couples conformer aggregation with protein context and conditions protein token encoding on ligand information.

- We introduce a protein-side Mamba residual refinement module that improves ordered protein representations while retaining explicit BiIntention-based drug-target matching.

We evaluate TAMR-DTI on BindingDB, BioSNAP, Human, and C. elegans under repeated-seed experimental protocols. Comparative results, module ablations, conformer-count controls, and conformer-weight diagnostics are used to examine both benchmark performance and the mechanism-level contribution of target-aware modulation and protein-only state-space refinement.

2 RELATED WORK

This section reviews the broader development of DTI prediction, beginning with classical similarity, matrix-factorization, and network-based methods before covering modern drug and protein representation learning with emphasis on recent 2023–2026 advances, and finally discusses interaction modeling and multimodal systems that motivate target-aware multimodal design.

2.1 PRE-DEEP-LEARNING DTI METHODS

Early DTI methods cover three families that remain conceptually relevant. Chemogenomic and similarity-inference approaches framed DTI as a link-prediction problem over drug and target spaces, exploiting the heuristic that chemically similar drugs bind functionally related proteins (Yamanishi et al., 2008; Keiser et al., 2007). Kernel learning and matrix-factorization variants formalized this view by recovering latent drug and target factors from sparse interaction matrices, regularized by neighborhood or profile similarity (Liu et al., 2016; He et al., 2017). Heterogeneous-network methods then extended pairwise similarity to broader biomedical relations spanning drugs, targets, diseases, side effects, and knowledge graphs (Luo et al., 2017; Wan et al., 2019; Ye et al., 2021). Classical machine-learning predictors, including support vector machines, random forests, nearest-neighbor classifiers, logistic regression, and boosting, served as strong baselines whenever molecular descriptors, protein features, and curated negatives were available (Cortes & Vapnik, 1995; Breiman, 2001; Guo et al., 2003; Friedman et al., 2000; Liu et al., 2015). These methods are interpretable and data-efficient but rely on curated similarity definitions and coarse pairwise summaries, which limits their ability to capture residue-level patterns, pair-specific molecular substructures, or three-dimensional binding geometry. Recent representation-learning frameworks continue to use these limitations as the motivation for cold-start, transfer, and mechanism-aware DTI modeling (Luo et al., 2024; Lu et al., 2025).

2.2 DRUG REPRESENTATION

Modern small-molecule encoders span the 1D, 2D, and 3D structural views of a ligand, a taxonomy also emphasized in recent reviews of molecular representation foundation models (Song et al., 2026). The one-dimensional view treats a molecule as a token sequence: SMILES strings provide a compact line notation for molecular graphs (Weininger, 1988), molecular fingerprints such as ECFP encode local substructure patterns into fixed-length vectors used in classical predictors and similarity search (Rogers & Hahn, 2010), and broader cheminformatics descriptors summarize physicochemical and topological properties (Todeschini & Consonni, 2009). Pretrained molecular language models extend the 1D view through self-supervised SMILES modeling: ChemBERTa, for example, learns transferable chemistry features from large molecule corpora (Chithrananda et al., 2020). Recent DTI models still show the value of this view when it is paired with stronger pretraining or split-aware representation controls, as in DLM-DTI and BarlowDTI (Lee et al., 2024; Schuh et al., 2025).

Two-dimensional graph encoders replace linear notations with explicit atom-bond topology. Neural graph fingerprints and message-passing networks made molecular representation learning differentiable and end-to-end (Duvenaud et al., 2015; Gilmer et al., 2017), and DTI-specific graph models such as GraphDTA, DGraphDTA, and MGraphDTA exploit molecular topology, protein contact maps, or multiscale graph representations to better capture chemical neighborhoods and their relation to binding (Nguyen et al., 2021; Jiang et al., 2020; Yang et al., 2022). Recent graph pretraining and DTI systems extend this trajectory: Mole-BERT rethinks molecular graph pretraining for transferable atom-level representations, while GEFormerDTA uses graph-structured atom features with Transformer-based early fusion for affinity prediction (Xia et al., 2023; Liu et al., 2024).

Three-dimensional encoders introduce binding-relevant geometry. Continuous-filter networks such as SchNet model coordinate-dependent molecular signals (Schütt et al., 2017), and equivariant graph neural networks (EGNN) capture the rotational and translational structure of conformer atoms with low overhead (Satorras et al., 2021). Large-scale 3D molecular pretraining such as Uni-Mol further demonstrates that conformer-aware representations can improve molecular property, conformation, and protein-ligand spatial tasks (Zhou et al., 2023). Existing multimodal DTI pipelines that include a 3D branch typically aggregate candidate conformers as a drug-side property (Satorras et al., 2021;

Ji et al., 2026), so the same ligand contributes the same geometric summary regardless of which protein it is paired with — a design choice that we revisit in Section 3.

2.3 Protein Representation

Protein-side modeling has shifted from hand-crafted descriptors to pretrained sequence encoders. Traditional features such as pseudo-amino-acid composition, sequence-similarity scores from BLAST-derived profiles, and domain or functional annotations from Pfam and Gene Ontology provide biologically interpretable signals, although they tend to miss fine-grained residue dependencies that are not captured by fixed descriptors (Chou, 2001; Altschul et al., 1997; Finn et al., 2014; Ashburner et al., 2000; Dubchak et al., 1995). Deep learning then introduced learned encoders: convolutional and recurrent networks capture local residue motifs and short-range dependencies (Lee et al., 2019; Karimi et al., 2019), while Transformer-style architectures model longer-range residue interactions through self-attention (Chen et al., 2020). Protein language models broadened this direction by pretraining on large protein corpora: ProtTrans and ProtBERT learn contextual residue embeddings from self-supervised sequence modeling, and ESM shows that scaled unsupervised learning recovers structural and functional signals from sequence alone (Elnaggar et al., 2022; Rives et al., 2021). ESM-2 and ESMFold further demonstrate that scaled protein language models can support high-resolution structure prediction from sequence alone, reinforcing the value of sequence-derived protein representations even when 3D annotations are limited (Lin et al., 2023). Pair-level extensions further use protein language space as the modeling substrate itself: ConPLex aligns drug and protein embeddings in a shared contrastive space, and DLM-DTI combines dual language models with a teacher-student hint mechanism that injects target-aware signal into a smaller student (Singh et al., 2023; Lee et al., 2024). These backbones supply strong general-purpose target embeddings; they are nevertheless learned independently of the specific drug partner that will be considered downstream, which leaves pair-specific protein modulation to later stages of the pipeline.

2.4 Interaction Modeling

After per-modality encoders, a DTI model must decide compatibility between drug and target representations. The simplest design — concatenation of unimodal embeddings before an MLP head — is broadly compatible across encoders but offloads all cross-modal reasoning onto compressed global vectors. Attention modules sharpen this by aligning compound substructures with protein residues: TransformerCPI applies sequence self-attention to compound-protein pairs (Chen et al., 2020), MolTrans learns substructure-level interactions between drug and protein tokens with Transformer and convolutional modules (Huang et al., 2021), and IIFDTI separates pair-specific (interactive) and modality-specific (independent) features through an attention design (Cheng et al., 2022). Recent multimodal variants such as MCL-DTI combine drug molecular images and chemical text with bidirectional cross-attention, showing that interaction modules are increasingly coupled with richer drug-side representations (Qian et al., 2023). Bilinear and bidirectional models raise the resolution of cross-modal matching further: DrugBAN uses bilinear attention to capture interpretable drug-target associations and add domain adaptation across distributions (Bai et al., 2023), and BINDTI combines a convolution-attention hybrid (ACmix) protein encoder with bidirectional multi-head attention that explicitly models drug-to-protein and protein-to-drug intention streams (Peng et al., 2025). Cross-attention frameworks such as MFCADTI integrate multiple feature views before prediction (Quan et al., 2025), while interaction-fingerprint methods rethink how pairwise evidence is represented: PSICHIC learns hierarchical protein-ligand interaction fingerprints that incorporate physicochemical channel information (Koh et al., 2024), and BarlowDTI uses a Barlow-Twins-style self-supervised objective to learn modern one-dimensional DTI representations under stronger split and representation controls (Schuh et al., 2025). Attention-based interaction modules have therefore moved DTI prediction beyond late concatenation; most current designs still perform pair-specific reasoning only after independent unimodal encoders have compressed their inputs, however, which can discard substructure-level or residue-level evidence that a more deeply coupled design would preserve.

2.5 Multimodal DTI Systems and Method Positioning

Multimodal frameworks combine the representational threads above rather than choose a single one. LDM-DTI integrates pretrained drug and protein language features with graph topology, equivariant geometric encoders, and a dynamic bidirectional interaction module (Ji et al., 2026). DrugLAMP couples pretrained biochemical representations with a multi-attention pooling head designed for transferable DTI prediction across data distributions (Luo et al., 2024). GEFormerDTA fuses graph-structured atom features with a Transformer encoder through early-fusion blocks for binding-affinity prediction (Liu et al., 2024). EviDTI adds an evidential learning output head on top of a multimodal DTI backbone to quantify predictive uncertainty alongside binary prediction (Zhao et al., 2025). DTIAM unifies DTI, affinity, and activation/inhibition mechanism prediction through self-supervised drug and target pretraining (Lu et al., 2025), MMDG-DTI emphasizes multimodal feature fusion under domain generalization (Hua et al., 2025), and TriDTI extends multimodal alignment to structural, sequential, and relational views for both drugs and proteins (Yun et al., 2026). Beyond pretrained graph models, the rapid progress of protein structure prediction has further made predicted three-dimensional protein structure an accessible resource for downstream target modeling (Jumper et al., 2021), although the present work focuses on sequence-derived protein representations to remain comparable with existing BiIntention-style benchmarks.

BiMA-DTI is especially relevant to state-space modeling in DTI because it explores a Mamba-Attention hybrid in which Mamba and self-attention blocks alternate to model long-range dependencies on both drug and protein streams (Shui et al., 2025). This line of work raises two design questions for multimodal DTI. One concerns whether state-space refinement should be applied to both drug and protein tokens, since drug tokens do not necessarily carry the same biologically meaningful linear order as protein residues. The other concerns how long-range sequence refinement should be combined with explicit cross-modal matching, because blending state-space dynamics and attention into a single fusion stage may obscure the bidirectional drug-target alignment bias that has proven useful in recent interaction models. In the main benchmark comparison in Section 4, GraphDTA, DrugBAN, IIFDTI, BINDTI, TransformerCPI, MolTrans, and LDM-DTI are used as the closest published baselines for which protocols, splits, and reported metrics are compatible with the repeated-seed evaluation; the other methods discussed above use different splits, evaluation protocols, or unavailable training code, which prevents like-for-like reproduction under the same protocol and limits them to contextual discussion.

3 Methodology

3.1 Problem Definition

We formulate drug-target interaction (DTI) prediction as a supervised binary classification problem. Given a drug molecule D, a target protein T, and a label $y \in \{0, 1\}$ indicating whether the pair interacts, the model learns a scoring function f_θ that estimates the interaction probability:

$$s = f_\theta(D, T), \qquad p = \sigma(s) = p_\theta(y = 1 \mid D, T), \qquad \hat{y} = \mathbf{1}[p \geq \tau], \tag{1}$$

where s is the prediction logit, $\sigma(\cdot)$ is the sigmoid function, $p \in [0, 1]$ is the predicted binding probability, τ is a validation-selected decision threshold, $\hat{y} \in \{0, 1\}$ is the predicted binary label, and θ denotes all trainable parameters. For a mini-batch $\mathcal{B} = \{(D_i, T_i, y_i)\}_{i=1}^{B}$, the learning objective is to assign larger logits to interacting pairs and smaller logits to non-interacting pairs.

Each drug is described by three complementary views. The first view is a SMILES-level molecular language representation extracted from ChemBERTa (Chithrananda et al., 2020). The second view is a two-dimensional molecular graph, which preserves atom-level topology and chemical bonds as in graph-based DTI models (Nguyen et al., 2021). The third view is a set of three-dimensional conformers encoded with an equivariant geometric network (Satorras et al., 2021). Each target protein is represented by residue-level ProtBERT features from the ProtTrans family (Elnaggar et al., 2022). The model must encode each modality, align drug geometry with the paired target, and perform explicit drug-protein matching.

Throughout this section, $d = 128$ denotes the shared hidden dimension. The drug graph contains at most $N_g = 290$ atom tokens, each conformer contains $N_c = 64$ resampled atom tokens, and the main model uses $K = 8$ conformers. The drug token length after topology-geometry concatenation is $L_d = N_g + N_c = 354$. The protein sequence length is denoted by L_p, with a binary mask $\mathbf{m} \in \{0, 1\}^{L_p}$ used to identify valid residues.

3.2 Drug Data Preprocessing

The preprocessing stage converts each SMILES string into three complementary drug views before neural encoding. The overview diagram summarizes this conversion in Figure 1. The 1D view captures molecular-language semantics: it is processed by ChemBERTa tokenization, frozen language-model encoding, and adaptive pooling so that pretrained chemical priors can enter the shared drug-token interface. The 2D view captures atom-bond topology: it is processed with RDKit/DGLLife graph construction, canonical atom and bond features, self-loops, and virtual-node padding so that local chemical connectivity is preserved. The 3D view captures plausible ligand geometry: it is processed through hydrogen addition, ETKDG conformer generation, optional UFF optimization, energy ranking, fixed-size atom-token pooling, and validity masking so that the downstream model can choose among multiple target-dependent conformations rather than relying on a single fixed geometry.

1D semantic preprocessing. The 1D module treats the SMILES string as a molecular language sequence. The string is tokenized and encoded by a frozen ChemBERTa-77M-MTR model (Chithrananda et al., 2020). Special tokens are removed, and the remaining token hidden states are adaptively pooled to the shared drug-token length and hidden dimension, giving $\mathbf{X}_{\text{chem}} \in \mathbb{R}^{354\times 128}$. This branch provides pretrained chemical semantics and remains independent of atom-index ordering in the graph and conformer branches.

2D topology preprocessing. The 2D module reconstructs the molecular graph from the same SMILES string using RDKit and DGLLife. Canonical atom and bond featurizers build a DGL molecular bigraph with self-loops, atom descriptors, bond descriptors, and adjacency structure. Atom features are augmented with a virtual-node indicator and padded with virtual nodes to $N_g = 290$ graph nodes, producing the graph input $\mathcal{G}_D$. This branch preserves local chemical connectivity and bond-induced neighborhood structure that are not explicit in the pooled ChemBERTa token matrix.

3D conformer preprocessing. The 3D module generates candidate ligand geometries. Hydrogens are first added, then RDKit ETKDGv3 embeds up to K conformers. When UFF parameters are available, each conformer is optimized and assigned a computed UFF energy. The candidate conformers are sorted by this energy, adaptively pooled to $N_c = 64$ atom tokens, and stored as atom features $\mathbf{F}_k$, Cartesian coordinates $\mathbf{R}_k$, normalized energy ϵ_k, and validity indicator q_k. Invalid or missing conformer slots are zero-padded and masked. This module therefore exposes multiple plausible geometries to the later target-aware conformer selector, rather than committing to a single fixed conformation during preprocessing.

Together, the three modules produce the cached drug-side inputs $\mathbf{X}_{\text{chem}}$, $\mathcal{G}_D$, and $\{\mathbf{F}_k, \mathbf{R}_k, \epsilon_k, q_k\}_{k=1}^{K}$ used by the drug encoder.

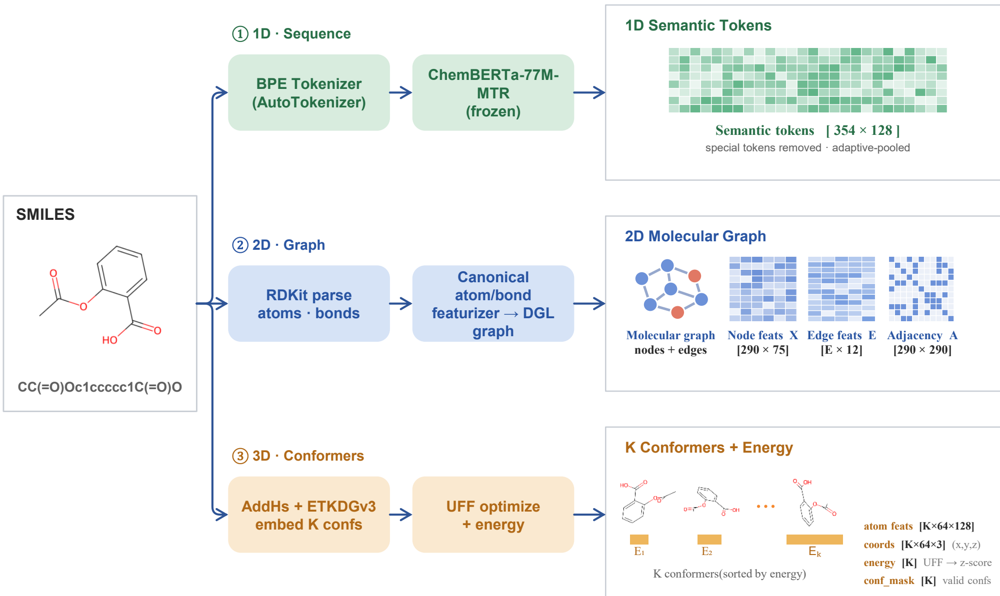

Figure 1: Drug data preprocessing. Starting from a SMILES string, TAMR-DTI constructs a ChemBERTa semantic-token matrix, an RDKit/DGL molecular graph, and a set of energy-ranked 3D conformers with atom features, coordinates, energies, and conformer-validity indicators.

3.3 Proposed Representation Framework

The proposed TAMR-DTI representation framework is built on a BiIntention-based DTI backbone (Peng et al., 2025; Ji et al., 2026). It is organized into six modules that are detailed in the following module descriptions. The drug representation module constructs a fused ligand token sequence from ChemBERTa semantics, graph topology, and target-aware conformer geometry. The protein representation module maps ProtBERT residue features into ligand-conditioned protein tokens through FiLM-modulated ACmix blocks. The aligned representation interface standardizes hidden dimensions, token lengths, context vectors, and masks before cross-modal comparison. The protein representation refinement and interaction module applies gated bidirectional Mamba only to the ordered protein token stream and then uses BiIntention for explicit drug-target matching (Gu et al., 2022; Gu & Dao, 2023). The prediction module converts the resulting interaction representation into a binding logit, and the training objective optimizes this logit with weighted binary cross-entropy.

The schematic overview in Figure 2 and the formal flow in Eq. 2 summarize the end-to-end representation path. The drug representation module receives the ChemBERTa semantic tokens, the molecular graph, the conformer cache, and the initial protein seed representation. It extracts topology and target-conditioned geometry, then fuses the drug-side streams into the token sequence $\mathbf{D}$. The fused drug tokens are pooled into the ligand context $\mathbf{x}_D$, which conditions the protein representation module. A gated bidirectional Mamba residual block refines the ligand-conditioned protein tokens, BiIntention forms the compact interaction representation, and the prediction module maps this representation to the interaction probability.

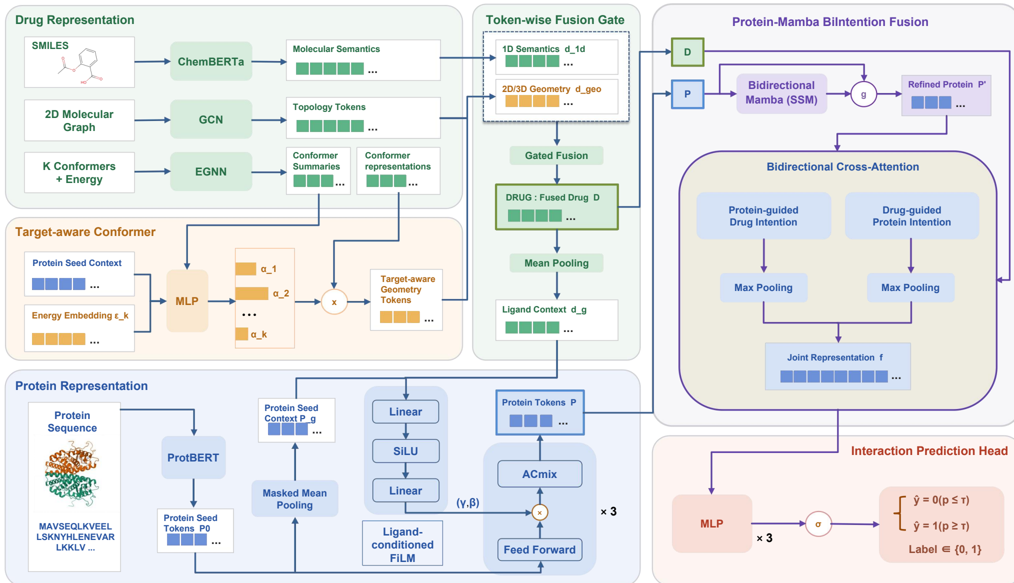

Figure 2: TAMR-DTI representation framework. The figure is a schematic view of the drug representation streams, ligand-conditioned protein representation module, protein representation refinement block, and BiIntention interaction representation head. Formal tensor notation is defined in Eq. 2 and the surrounding text. Glyphs: $\times$ multiplication, $\odot$ element-wise product, σ sigmoid, and "$\times 3$" a block stacked three times.

The computation is

$$\begin{aligned}
\mathbf{D} &= \mathrm{DrugEnc}\big(\mathbf{X}_{\text{chem}}, \mathcal{G}_D, \{\mathbf{F}_k, \mathbf{R}_k, \epsilon_k, q_k\}_{k=1}^{K}, \mathbf{P}^0, \mathbf{m}\big)\,, \\
\mathbf{x}_D &= \mathrm{MeanPool}(\mathbf{D}), \\
\mathbf{P} &= \mathrm{ProtEnc}\big(\mathbf{P}^0, \mathbf{m}, \mathbf{x}_D\big)\,, \\
\mathbf{P}' &= \mathrm{ProtMamba}(\mathbf{P}, \mathbf{m}), \\
\mathbf{f} &= \mathrm{BiIntention}(\mathbf{D}, \mathbf{P}'), \\
s &= \mathrm{MLP}(\mathbf{f}), \qquad p = \sigma(s).
\end{aligned} \tag{2}$$

Here $\mathbf{X}_{\text{chem}} \in \mathbb{R}^{L_d \times d}$ denotes the ChemBERTa drug feature sequence, $\mathcal{G}_D$ is the molecular graph, $(\mathbf{F}_k, \mathbf{R}_k)$ are the atom features and coordinates of conformer k, ϵ_k is its conformer energy, and $q_k \in \{0, 1\}$ marks whether the conformer is valid. $\mathbf{P}^0$ is the projected ProtBERT protein feature, $\mathbf{D} \in \mathbb{R}^{L_d \times d}$ is the fused drug token sequence, $\mathbf{P} \in \mathbb{R}^{L_p \times d}$ is the ligand-conditioned protein token sequence before protein-side Mamba refinement, $\mathbf{P}'$ is the refined protein token sequence, $\mathbf{x}_D$ is the global ligand context used for ligand-conditioned protein encoding, and $\mathbf{f} \in \mathbb{R}^{2d}$ is the fused pair representation. The projected protein sequence first guides conformer aggregation, and the resulting drug representation then provides ligand context for protein encoding before final cross-modal matching.

3.3.1 Drug Representation Module

The drug representation module integrates molecular language, graph topology, and target-aware 3D geometry. SMILES features encode chemical semantics learned from molecular corpora, graph features preserve atom-neighbor connectivity, and conformer features encode spatial arrangement. TAMR-DTI first represents the one-dimensional SMILES stream with a frozen molecular language model, computes topology and geometry tokens from RDKit-derived structures, and then fuses the three drug views through a learnable gate.

1D semantic representation. The 1D branch starts from the SMILES string s_D and treats it as a molecular language sequence. During preprocessing, s_D is tokenized with the ChemBERTa tokenizer and encoded by the frozen ChemBERTa-77M-MTR backbone:

$$\mathbf{U}^{\text{chem}} = \text{ChemBERTa}(s_D) \in \mathbb{R}^{L_s \times d_c}. \tag{3}$$

Here L_s and d_c denote the ChemBERTa sequence length and hidden size before fixed-size pooling. Special tokens are removed, and the remaining token hidden states are pooled to the shared drug-token length and hidden dimension:

$$\mathbf{X}_{\text{chem}} = \text{Pool}_{L_d,d}\left(\mathbf{U}^{\text{chem}}_{\neg\text{special}}\right) \in \mathbb{R}^{L_d \times d}. \tag{4}$$

The resulting sequence provides pretrained SMILES semantics. Its positions are fixed molecular-language slots after pooling, not atom indices, which is why the later fusion gate is interpreted as a latent-slot fusion rather than a direct SMILES-token-to-atom alignment.

2D topology representation. Let $\mathcal{G}_D = (\mathcal{V}, \mathcal{E})$ be the padded molecular graph of drug D, where each atom node has an input feature $\mathbf{a}_i \in \mathbb{R}^{75}$. The implementation first projects atom features to the hidden space:

$$\mathbf{h}_i^{g,0} = \mathbf{W}_g \mathbf{a}_i, \qquad \mathbf{H}^{g,0} \in \mathbb{R}^{N_g \times d}. \tag{5}$$

When graph padding is enabled, the last input channel is used as the virtual padding marker, and its projection weight is fixed to zero so that padded nodes do not introduce artificial atom semantics. A multi-layer graph convolutional network then aggregates local chemical topology:

$$\mathbf{H}^g = \text{GCN}(\mathcal{G}_D, \mathbf{H}^{g,0}) \in \mathbb{R}^{N_g \times d}. \tag{6}$$

The resulting $\mathbf{H}^g$ provides atom-level topology tokens.

3D conformer representation. For conformer k, let $\mathbf{F}_k \in \mathbb{R}^{N_c \times d}$ denote atom features and $\mathbf{R}_k \in \mathbb{R}^{N_c \times 3}$ denote Cartesian coordinates. TAMR-DTI uses an EGNN layer to update atom features while respecting Euclidean symmetry:

$$(\mathbf{Z}_k, \mathbf{R}'_k) = \text{EGNN}(\mathbf{F}_k, \mathbf{R}_k), \qquad \mathbf{Z}_k \in \mathbb{R}^{N_c \times d}. \tag{7}$$

The conformer-level summary is obtained by average pooling over the fixed resampled atom tokens:

$$\mathbf{c}_k = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{z}_{k,i}. \tag{8}$$

This summary is used only to score conformer relevance; the final 3D drug representation remains atom-resolved.

The conformer slots are kept in the order produced by preprocessing. RDKit ETKDG is used to generate candidate conformers, UFF optimization is applied when force-field parameters are available, and the resulting conformers are sorted by their computed UFF energy before padding or truncation. Therefore, the first slot corresponds to the lowest computed-energy conformer among the generated candidates, slots two to four provide lower-energy alternative conformations, and later slots contain higher-energy generated alternatives. The energy value is used as an auxiliary conformer-ranking feature rather than as a calibrated cross-molecule thermodynamic quantity. The conformer-count comparison follows this order: the $n = 1$ setting tests a fixed lowest-computed-energy geometry, whereas larger n values expose progressively broader geometric alternatives to the target-aware selector.

Target-aware conformer representation. Conformer aggregation depends on the paired protein. The same ligand can expose different binding-relevant geometric cues for different targets, so a fixed conformer average may dilute target-specific 3D signal. ProtBERT tokens are first normalized and projected:

$$\mathbf{P}^0 = \mathbf{W}_p \ \text{LayerNorm}(\mathbf{P}_{\text{ProtBERT}}), \qquad \mathbf{P}_{\text{ProtBERT}} \in \mathbb{R}^{L_p \times 1024}, \quad \mathbf{P}^0 \in \mathbb{R}^{L_p \times d}. \tag{9}$$

The masked protein context vector is then

$$\mathbf{x}_T = \frac{\sum_{j=1}^{L_p} m_j \mathbf{p}_j^0}{\max(1, \sum_{j=1}^{L_p} m_j)}. \tag{10}$$

For each conformer, its energy ϵ_k is projected to the hidden space, $\mathbf{u}_k = \mathbf{W}_e \epsilon_k$. The conformer logit combines the conformer summary, the protein context, and the energy embedding:

$$a_k = \mathbf{w}_a^\top \, \mathrm{SiLU}\left(\mathbf{W}_a[\mathbf{c}_k; \mathbf{x}_T; \mathbf{u}_k] + \mathbf{b}_a\right) + b_s. \tag{11}$$

Invalid conformers are masked by assigning their logits a large negative value. For samples with at least one valid conformer, the conformer weights are

$$\alpha_k = \frac{\exp(a_k)\, q_k}{\sum_{\ell=1}^{K} \exp(a_\ell)\, q_\ell}, \qquad \sum_{k=1}^{K} \alpha_k = 1. \tag{12}$$

In the rare case that preprocessing yields an empty conformer-validity mask, the implementation activates the first conformer slot as a fallback before applying the softmax, preventing a zero denominator in Eq. 12. The target-aware atom-level 3D representation and its global context are

$$\mathbf{Z}^{3D} = \sum_{k=1}^{K} \alpha_k \mathbf{Z}_k, \qquad \mathbf{c}^{3D} = \sum_{k=1}^{K} \alpha_k \mathbf{c}_k. \tag{13}$$

Thus, conformer weights are conditioned on the paired protein rather than fixed for the ligand alone.

Drug representation fusion. The graph topology tokens and target-aware 3D tokens are concatenated:

$$\mathbf{X}_{\mathrm{str}} = [\mathbf{H}^g; \mathbf{Z}^{3D}] \in \mathbb{R}^{(N_g+N_c)\times d}. \tag{14}$$

By construction, this structural sequence has length $N_g + N_c = L_d$ and matches the 1D ChemBERTa feature sequence $\mathbf{X}_{\mathrm{chem}} \in \mathbb{R}^{L_d \times d}$. Therefore, the fusion gate should be understood as a fixed latent-slot operation over two equal-length drug representations after preprocessing. The slot index does not imply a one-to-one chemical alignment between a SMILES token and a graph atom. Both branches are normalized and combined by a slot-wise, channel-wise gate:

$$\begin{aligned}
\tilde{\mathbf{X}}_{\mathrm{chem}} &= \mathrm{LayerNorm}(\mathbf{X}_{\mathrm{chem}}), \\
\tilde{\mathbf{X}}_{\mathrm{str}} &= \mathrm{LayerNorm}(\mathbf{X}_{\mathrm{str}}), \\
\mathbf{G}^D &= \sigma\Big(\mathbf{W}_f[\tilde{\mathbf{X}}_{\mathrm{chem}}; \tilde{\mathbf{X}}_{\mathrm{str}}] + \mathbf{b}_f\Big), \\
\mathbf{D} &= \mathbf{G}^D \odot \tilde{\mathbf{X}}_{\mathrm{chem}} + (1 - \mathbf{G}^D) \odot \tilde{\mathbf{X}}_{\mathrm{str}}.
\end{aligned} \tag{15}$$

The gate bias is initialized to -2.0, so the initial value $\sigma(-2.0) \approx 0.119$ assigns a small but nonzero weight to the ChemBERTa branch and a larger initial weight to the topology-geometry branch. During training, the gate learns the token-wise balance between pretrained molecular semantics and structural information. The global ligand context used by the protein module is

$$\mathbf{x}_D = \frac{1}{L_d} \sum_{i=1}^{L_d} \mathbf{d}_i. \tag{16}$$

3.3.2 Protein Representation Module

The protein representation module maps ProtBERT residue features into ligand-conditioned protein tokens. ProtBERT provides residue-level sequence priors, and the Feature-wise Linear Modulation (FiLM) path (Perez et al., 2018) lets the paired ligand modulate these tokens before cross-modal fusion.

The input protein sequence is first mapped from the ProtBERT dimension to the shared hidden dimension using Eq. 9. This is the same protein seed sequence used by the drug representation module for target-aware conformer selection. The projected tokens $\mathbf{P}^0$ are then passed through three ProteinACmix layers. Each layer combines a feed-forward half-step, ligand-conditioned feature modulation, relative self-attention, dynamic convolution, and a second feed-forward half-step. For layer $\ell = 1, \ldots, 3$, let $\mathbf{P}^{\ell-1} \in \mathbb{R}^{L_p \times d}$ be the input tokens. The first feed-forward residual update is

$$\mathbf{U}^\ell = \frac{1}{2}\, \mathrm{FFN}_1^\ell(\mathbf{P}^{\ell-1}) + \frac{1}{2}\mathbf{P}^{\ell-1}. \tag{17}$$

If ligand conditioning is enabled, the global drug context $\mathbf{x}_D$ is converted into feature-wise scale and shift parameters $\boldsymbol{\gamma}^\ell, \boldsymbol{\beta}^\ell \in \mathbb{R}^d$:

$$\begin{aligned} \left[\boldsymbol{\gamma}^\ell, \boldsymbol{\beta}^\ell\right] &= \mathrm{MLP}^\ell_{film}(\mathbf{x}_D), \\ \bar{\boldsymbol{\gamma}}^\ell &= \rho \tanh(\boldsymbol{\gamma}^\ell), \qquad\qquad \rho = 0.05. \\ \bar{\boldsymbol{\beta}}^\ell &= \rho \boldsymbol{\beta}^\ell, \end{aligned} \tag{18}$$

The protein tokens are then modulated as

$$\tilde{\mathbf{U}}^\ell = \mathbf{U}^\ell \odot (1 + \bar{\boldsymbol{\gamma}}^\ell) + \bar{\boldsymbol{\beta}}^\ell. \tag{19}$$

where the scale and shift vectors are broadcast over the residue-token axis. When ligand conditioning is disabled, the implementation uses $\tilde{\mathbf{U}}^\ell = \mathbf{U}^\ell$. The fixed scale ρ keeps ligand modulation small at initialization.

The modulated tokens are further processed by the CNN-Transformer block:

$$\begin{aligned} \mathbf{A}^\ell &= \mathrm{MHSA}^\ell_{rel}(\tilde{\mathbf{U}}^\ell, \mathbf{m}) + \tilde{\mathbf{U}}^\ell, \\ \mathbf{C}^\ell &= \mathrm{Conv}^\ell(\mathbf{A}^\ell) + \mathbf{A}^\ell, \\ \mathbf{P}^\ell &= \frac{1}{2}\,\mathrm{FFN}^\ell_2(\mathbf{C}^\ell) + \frac{1}{2}\mathbf{C}^\ell. \end{aligned} \tag{20}$$

The relative multi-head self-attention captures long-range residue dependencies under the protein mask, and the convolutional branch emphasizes local sequence motifs that may be relevant to interaction prediction. After three layers, the protein module outputs

$$\mathbf{P} = \mathbf{P}^3 \in \mathbb{R}^{L_p \times d}. \tag{21}$$

3.3.3 Aligned Representation Interface

The preceding representation modules operate on heterogeneous inputs: cached ChemBERTa tokens, RDKit atom descriptors, EGNN conformer tokens, and ProtBERT residue embeddings. Before the interaction module can compare drug and protein tokens, TAMR-DTI converts these streams into a shared tensor interface with a common hidden dimension, fixed drug-token length, and explicit validity masks. This interface is an implementation contract between the modality-specific representation modules and the interaction module.

The active-stream projections used by the encoders are summarized as mappings to the shared dimension $d = 128$:

$$\psi_{\text{chem}} : \mathbb{R}^{128} \to \mathbb{R}^{128}, \qquad \psi_g : \mathbb{R}^{75} \to \mathbb{R}^{128}, \qquad \psi_p : \mathbb{R}^{1024} \to \mathbb{R}^{128}. \tag{22}$$

Here ψ_{chem} is the layer normalization applied to the cached ChemBERTa sequence, ψ_g is the graph input projection in Eq. 5, and ψ_p is the LayerNorm-linear projection of ProtBERT features in Eq. 9. The conformer atom features are prepared in the same hidden dimension and then updated by the EGNN. After the GCN and target-aware conformer branches, the structural drug sequence has length $N_g + N_c = 290 + 64 = 354$, matching the cached ChemBERTa sequence. The drug fusion gate therefore acts on paired latent slots of equal length, rather than on an assumed one-to-one correspondence between SMILES tokens and graph atoms.

Second, TAMR-DTI constructs pair-level contexts in a fixed order. The projected protein seed tokens are pooled with the residue mask to form the target context used for conformer selection. After the target-aware drug tokens are fused, their unmasked mean becomes the ligand context used by FiLM in the protein representation module:

$$\mathbf{x}_T = \mathrm{MaskedMean}(\mathbf{P}^0, \mathbf{m}), \qquad \mathbf{x}_D = \mathrm{MeanPool}(\mathbf{D}). \tag{23}$$

This ordering implements the two target-aware paths in the model: protein context first selects ligand conformers, and the resulting ligand context then modulates protein residues.

Third, masks are applied only at stages where the corresponding operation is mask-aware in the implementation. The conformer validity mask $\mathbf{q} = (q_1, \ldots, q_K)$ prevents invalid conformers from receiving probability mass in Eq. 12. The protein mask $\mathbf{m}$ is used in masked mean pooling, relative self-attention inside the protein representation module, and protein-side Mamba refinement. The final BiIntention module then receives the fixed-length protein token sequence produced by these masked upstream operations. Thus, the aligned interface entering the fusion stage is $\{\mathbf{D}, \mathbf{P}, \mathbf{m}\}$, while $\boldsymbol{\alpha}$ is retained as an auxiliary record of the target-aware conformer weights for analysis.

3.3.4 Protein Representation Refinement and Interaction Module

The protein representation refinement and interaction module combines protein-side sequence refinement with BiIntention-based drug-target matching. The main configuration uses `protein_mamba_biintention`: Mamba is applied only to the protein token stream, and BiIntention remains the pairwise interaction module. Protein residues form an ordered biological sequence, whereas drug tokens concatenate SMILES semantics, graph topology, and conformer geometry.

Protein representation refinement. Given protein tokens $\mathbf{P}$ and mask $\mathbf{m}$, padded positions are suppressed before entering the Mamba residual block:

$$\mathbf{P}_j^{mask} = m_j \mathbf{p}_j. \tag{24}$$

The Mamba residual block normalizes the sequence, applies a forward selective state-space mixer, and applies a backward mixer to the reversed sequence:

$$\begin{aligned}
\mathbf{H} &= \mathrm{LayerNorm}(\mathbf{P}^{mask}), \\
\mathbf{M}_f &= \mathrm{Mamba}_f(\mathbf{H}), \\
\mathbf{M}_b &= \mathrm{Reverse}(\mathrm{Mamba}_b(\mathrm{Reverse}(\mathbf{H}))), \\
\mathbf{M} &= \mathbf{W}_m[\mathbf{M}_f; \mathbf{M}_b].
\end{aligned} \tag{25}$$

The block then uses residual and feed-forward updates:

$$\begin{aligned}
\mathbf{R} &= \mathbf{P}^{mask} + \mathrm{Dropout}(\mathbf{M}), \\
\bar{\mathbf{P}}_{\mathrm{Mamba}} &= \mathbf{R} + \mathrm{FFN}_m(\mathbf{R}), \\
\mathbf{P}_{\mathrm{Mamba},j} &= m_j \bar{\mathbf{p}}_{\mathrm{Mamba},j}.
\end{aligned} \tag{26}$$

A scalar gate controls how much of the refined protein representation is injected:

$$g = \sigma(\eta), \qquad \mathbf{P}' = \mathbf{P} + g\,(\mathbf{P}_{\mathrm{Mamba}} - \mathbf{P}). \tag{27}$$

The gate logit is initialized as $\eta = -2.0$, giving $g \approx 0.119$. The mask is applied inside the Mamba residual block; the subsequent BiIntention module receives the resulting fixed-length protein token sequence.

Bidirectional interaction representation. After protein-side refinement, the bidirectional interaction representation path uses BiIntention to model explicit drug-protein compatibility. Self-attention is first applied independently to each modality:

$$\mathbf{D}^s = \mathrm{SA}_D(\mathbf{D}), \qquad \mathbf{P}^s = \mathrm{SA}_P(\mathbf{P}'), \tag{28}$$

where each self-attention module includes a residual connection. The intention operator maps a source sequence $\mathbf{X} \in \mathbb{R}^{n_x \times d}$ and a query sequence $\mathbf{Q} \in \mathbb{R}^{n_q \times d}$ to a compact cross-modal intention representation. We set $\mathbf{D}^{(0)} = \mathbf{D}^s$ and $\mathbf{P}^{(0)} = \mathbf{P}^s$ for the following bidirectional update. For attention head r with head dimension $d_h = d/H$,

$$\begin{aligned}
\mathbf{Q}_r &= \mathbf{Q}\mathbf{W}_r^Q, \\
\mathbf{K}_r &= \mathbf{X}\mathbf{W}_r^K, \\
\mathbf{V}_r &= \mathbf{X}\mathbf{W}_r^V, \\
\mathbf{A}_r &= \omega_r \ \mathrm{softmax}_{\mathrm{feat}}\left((\alpha_{\mathrm{ridge}}\mathbf{I} + \mathbf{K}_r^\top \mathbf{K}_r)^\dagger \mathbf{K}_r^\top \mathbf{V}_r\right), \\
\mathbf{O}_r &= \mathbf{Q}_r \mathbf{A}_r,
\end{aligned} \tag{29}$$

where α_{ridge} is a shared learnable ridge parameter, ω_r is a learnable head weight, $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse, and $\mathrm{softmax}_{\mathrm{feat}}$ follows the implementation's head-feature axis. Concatenating heads and applying an output projection gives $\mathrm{Int}(\mathbf{X}, \mathbf{Q}) \in \mathbb{R}^{n_q \times d}$. The intention block then computes

$$\Phi(\mathbf{X}, \mathbf{Q}) = \beta\,\mathbf{Q}^\top \mathrm{softmax}\left(\mathrm{Int}(\mathrm{LayerNorm}(\mathbf{X}), \mathbf{Q})\right), \tag{30}$$

with learnable scalar β, yielding a tensor in $\mathbb{R}^{d\times d}$. For the ℓ-th BiIntention layer,

$$\begin{aligned}\mathbf{V}_p^{(\ell)} &= \Phi(\mathbf{D}^{(\ell-1)}, \mathbf{P}^{(\ell-1)}),\\ \mathbf{V}_d^{(\ell)} &= \Phi(\mathbf{P}^{(\ell-1)}, \mathbf{D}^{(\ell-1)}),\\ \mathbf{D}^{(\ell)} &= \mathbf{V}_d^{(\ell)}, \qquad \mathbf{P}^{(\ell)} = \mathbf{V}_p^{(\ell)}.\end{aligned} \tag{31}$$

The main model uses one BiIntention layer and eight attention heads. The final BiIntention tensors are not used as replacement unimodal encodings; instead, they are compact bidirectional interaction representations from which the drug-side and protein-side pair features are pooled. The implementation applies max pooling over the final fixed-length BiIntention tensors:

$$\mathbf{f}_d = \mathrm{MaxPool}(\mathbf{D}^{(L)}), \qquad \mathbf{f}_p = \mathrm{MaxPool}(\mathbf{P}^{(L)}), \qquad \mathbf{f} = [\mathbf{f}_d; \mathbf{f}_p] \in \mathbb{R}^{2d}. \tag{32}$$

3.3.5 Prediction Module

The prediction head maps the fused pair representation $\mathbf{f} \in \mathbb{R}^{256}$ to a scalar interaction logit. It is implemented as a four-layer MLP with layer normalization after the first three hidden transformations:

$$\begin{aligned}\mathbf{z}_1 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_1\mathbf{f} + \mathbf{b}_1)\right), \quad \mathbf{z}_1 \in \mathbb{R}^{256},\\ \mathbf{z}_2 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_2\mathbf{z}_1 + \mathbf{b}_2)\right), \quad \mathbf{z}_2 \in \mathbb{R}^{256},\\ \mathbf{z}_3 &= \mathrm{LayerNorm}\left(\mathrm{ReLU}(\mathbf{W}_3\mathbf{z}_2 + \mathbf{b}_3)\right), \quad \mathbf{z}_3 \in \mathbb{R}^{128},\\ s &= \mathbf{W}_4\mathbf{z}_3 + \mathbf{b}_4.\end{aligned} \tag{33}$$

The predicted probability is obtained by applying the sigmoid function as in Eq. 1.

3.3.6 Training Objective

The active training objective is weighted binary cross-entropy with logits. For sample i, let s_i be the model logit and $p_i = \sigma(s_i)$ the corresponding predicted probability. The mini-batch loss is

$$\mathcal{L}_{BCE} = -\frac{1}{B}\sum_{i=1}^{B}\left[w_+ y_i \log p_i + (1-y_i)\log(1-p_i)\right], \tag{34}$$

where w_+ is the positive-class weight. When automatic positive weighting is enabled, the weight is computed from the training split as

$$w_+ = \frac{N_-}{N_+}, \tag{35}$$

where N_+ and N_- are the numbers of positive and negative training samples.

All reported main experiments optimize Eq. 34 directly. Checkpoint selection, optimizer settings, learning-rate scheduling, early stopping, and threshold conventions are reported in Section 4.1.2. In particular, the main results select checkpoints by validation AUROC; fixed-threshold metrics are reported as Acc@0.5 and F1@0.5, while validation-selected operating points, when used, are reported separately as Acc(val-thr) and F1(val-thr).

4 Experiments

4.1 Experimental Settings

This subsection defines the experimental protocol used throughout the main benchmark study. It first introduces the four processed DTI datasets and their label distributions, then specifies the training configuration and class-weight settings used for repeated runs, and finally defines the ranking and fixed-threshold metrics used to compare models. All main results use the full eight-conformer TAMR-DTI model and five fixed random seeds, $\{42, 43, 44, 45, 46\}$. For every run, the best checkpoint is selected by validation AUROC. When a threshold-dependent result uses a tuned operating point, the classification threshold is selected only on the validation split and is then applied unchanged to the held-out test split; such results are reported as Acc(val-thr) and F1(val-thr). Fixed-threshold results, when reported, are denoted Acc@0.5 and F1@0.5.

4.1.1 Datasets and Data Distribution

We evaluate TAMR-DTI on four benchmark drug-target interaction (DTI) datasets: BindingDB, BioSNAP, Human, and C. elegans. Each sample consists of a drug SMILES string, a protein sequence, and a binary interaction label. Table 1 reports the aggregate statistics after merging the processed training, validation, and test split files used in this study. These processed-split statistics are used throughout the experiments, because they reflect the exact drug-target pairs available to the model after preprocessing.

BindingDB is a public database of experimentally measured protein-ligand binding affinities and drug-target interactions (Liu et al., 2007). In our processed splits, it is the largest benchmark, with 49,199 labeled interactions, 14,643 drugs, and 2,623 proteins. Its positive rate is 42.02%, giving a moderate negative-label skew. We use BindingDB to test whether TAMR-DTI can scale to broad chemical and target coverage while remaining robust under non-balanced label distributions.

BioSNAP comes from the Stanford Biomedical Network Dataset Collection and provides curated biomedical network data, including drug-target associations (Zitnik et al., 2018). The processed BioSNAP benchmark is medium-sized, with 27,464 interactions, and is nearly class-balanced, with a 50.36% positive rate. Compared with BindingDB, it offers a different curation source and a more balanced decision setting. We therefore use BioSNAP to evaluate generalization under a network-derived DTI distribution and to support the component analyses in later sections.

The Human benchmark follows the compound-protein interaction dataset introduced with high-confidence negative sample construction (Liu et al., 2015). It is the smallest benchmark in our study, with 5,997 interactions and a 43.91% positive rate. Its compact benchmark size and stronger negative skew make it useful for testing model stability when fewer training examples are available and thresholded metrics are more sensitive to class distribution.

The C. elegans benchmark uses the corresponding Caenorhabditis elegans compound-protein interaction data from the same dataset family (Liu et al., 2015). The processed split contains 7,786 interactions and is exactly balanced in aggregate. Because it represents a non-human model organism, it complements the Human benchmark and broadens the evaluation of TAMR-DTI beyond human-derived interaction data.

Table 1: Dataset statistics used in the main experiments. Counts are computed from the processed training, validation, and test split files used in this study.

Dataset	Drugs	Proteins	Interactions	Positive	Negative	Positive rate
BindingDB	14,643	2,623	49,199	20,674	28,525	42.02%
BioSNAP	4,505	2,181	27,464	13,830	13,634	50.36%
Human	2,726	2,001	5,997	2,633	3,364	43.91%
C. elegans	1,767	1,876	7,786	3,893	3,893	50.00%

4.1.2 Experimental Configuration

The architecture of TAMR-DTI is described in Section 3; this subsection reports only the hardware, software, training, optimization, and evaluation settings used in the main repeated-seed experiments. The main runs were conducted on eight NVIDIA Tesla P100 GPUs using a Python 3.9 environment with PyTorch, DeepSpeed, DGL/DGLLife, RDKit, transformers, and mamba-ssm. All datasets use fixed random train/validation/test splits. The model is optimized with Adam and weighted binary cross-entropy. The basic training hyperparameters are listed in Table 2.

The learning rate follows a plateau decay strategy: if validation performance does not improve for five epochs, the learning rate is multiplied by 0.5 until reaching the minimum learning rate. Gradients are clipped with a maximum norm of 1.0 to reduce unstable updates. The reported runs are launched with DeepSpeed ZeRO-2 on eight GPUs, but distributed-system details are treated as runtime settings rather than core training hyperparameters. Checkpoints are selected by validation AUROC with a minimum improvement of 1×10^{-4}, and early stopping is triggered after ten validation epochs without sufficient improvement. When Acc(val-thr) or F1(val-thr) is reported, the threshold is searched only on the validation set over the grid $\{0.01, 0.02, \ldots, 0.99\}$ and then applied unchanged to the held-out test set.

Table 2: Basic training hyperparameters for the main experiments.

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Maximum epochs	100
Micro-batch size	16 per GPU
Gradient accumulation	4 steps
Gradient clipping	1.0
Learning-rate decay factor	0.5
Minimum learning rate	1×10^{-6}

The weighted binary cross-entropy loss is applied to the logit z_i for each drug-target pair:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[w_+ y_i \log \sigma(z_i) + (1 - y_i) \log \left(1 - \sigma(z_i)\right) \right], \tag{36}$$

where $y_i \in \{0, 1\}$ is the interaction label, $\sigma(\cdot)$ is the sigmoid function, and w_+ is the positive-class weight. When automatic positive weighting is enabled, the positive-class weight is computed separately for each dataset as

$$w_+ = \frac{N_-^{\text{train}}}{N_+^{\text{train}}}, \tag{37}$$

where N_+^{train} and N_-^{train} are the numbers of positive and negative samples in the training split. Table 3 reports the resulting values. This setting reduces the effect of class imbalance on BindingDB and Human while leaving the near-balanced BioSNAP and C. elegans datasets almost unchanged.

Table 3: Dataset-specific positive-class weights used by the binary cross-entropy loss.

Dataset	Training positives	Training negatives	Positive-class weight
BindingDB	14,583	19,856	1.3616
BioSNAP	9,684	9,540	0.9851
Human	1,846	2,351	1.2736
C. elegans	2,727	2,723	0.9985

4.1.3 Evaluation Metrics

We report two threshold-independent ranking metrics, AUROC and AUPRC, and two threshold-dependent classification metrics, Acc@0.5 and F1@0.5. For a binary DTI prediction task, true positives (TP) are interacting drug-target pairs correctly predicted as positive, true negatives (TN) are non-interacting pairs correctly predicted as negative, false positives (FP) are non-interacting pairs predicted as positive, and false negatives (FN) are interacting pairs predicted as negative.

AUROC is the area under the receiver operating characteristic (ROC) curve. For a decision threshold τ, the true positive rate and false positive rate are

$$\text{TPR}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \qquad \text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)}. \tag{38}$$

The ROC curve varies τ over the prediction-score range, so AUROC measures positive-negative ranking quality independent of any single threshold. AUROC therefore is not reported with an @0.5 suffix.

At a fixed operating threshold, sensitivity is equivalent to recall or true positive rate, while specificity measures the true negative rate:

$$\mathrm{Sensitivity}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FN}(\tau)}, \qquad \mathrm{Specificity}(\tau) = \frac{\mathrm{TN}(\tau)}{\mathrm{TN}(\tau) + \mathrm{FP}(\tau)}. \tag{39}$$

Sensitivity reflects how many true interactions are recovered, whereas specificity reflects how many non-interacting pairs are correctly rejected. Both metrics depend on the chosen threshold and should therefore be interpreted as operating-point metrics rather than ranking metrics.

AUPRC is the area under the precision-recall curve. For threshold τ,

$$\mathrm{Precision}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FP}(\tau)}, \qquad \mathrm{Recall}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FN}(\tau)}. \tag{40}$$

AUPRC emphasizes retrieval quality for the positive interaction class and is useful when positive-label rates differ across datasets. Like AUROC, AUPRC is computed across thresholds and is not an @0.5 metric.

For fixed-threshold classification metrics, we convert a predicted interaction probability p_i into a binary prediction using $\hat{y}_i^{0.5} = \mathbb{I}(p_i \geq 0.5)$. Acc@0.5 measures overall correctness at this fixed threshold,

$$\mathrm{Acc@0.5} = \frac{\mathrm{TP}_{0.5} + \mathrm{TN}_{0.5}}{\mathrm{TP}_{0.5} + \mathrm{TN}_{0.5} + \mathrm{FP}_{0.5} + \mathrm{FN}_{0.5}}, \tag{41}$$

where the subscript 0.5 indicates counts computed after thresholding at 0.5. F1@0.5 is the harmonic mean of precision and recall at the same fixed threshold,

$$\mathrm{F1@0.5} = \frac{2\,\mathrm{Precision}_{0.5}\,\mathrm{Recall}_{0.5}}{\mathrm{Precision}_{0.5} + \mathrm{Recall}_{0.5}} = \frac{2\,\mathrm{TP}_{0.5}}{2\,\mathrm{TP}_{0.5} + \mathrm{FP}_{0.5} + \mathrm{FN}_{0.5}}. \tag{42}$$

Thresholded accuracy and F1 must always be interpreted together with their threshold. If a result uses a validation-selected threshold τ_{val} instead of 0.5, we denote it as Acc(val-thr) or F1(val-thr). In that setting, τ_{val} is selected only on the validation split and then applied unchanged to the test split. For TAMR-DTI, repeated-run summaries are reported as mean $\pm$ standard deviation across five seeds. We also report 95% confidence intervals computed as $\bar{x} \pm t_{0.975,4} s/\sqrt{5}$, with $t_{0.975,4} = 2.776$.

4.2 Baseline Methods

SVM (Cortes & Vapnik, 1995) is included as a reported classical machine-learning baseline. We adopt its values, together with all other baseline values in this section, from the reported comparison of Ji et al. (2026) rather than reimplementing or retraining the baseline methods. SVM provides a maximum-margin classifier for separating drug-target feature vectors with a high-dimensional decision boundary. In the DTI setting, it serves as a non-neural reference for whether fixed pairwise features are already separable without learned molecular or protein encoders.

RF (Breiman, 2001) is included as a reported classical machine-learning baseline based on tree ensembles. It aggregates multiple decision trees trained on bootstrapped samples and feature subsets, with the reported comparison using 100 trees. This baseline can capture nonlinear feature interactions without gradient-based representation learning, making it a useful classical reference against graph and attention models. Its inclusion also helps separate gains from learned interaction modeling from gains that may come from strong tabular-style feature partitioning.

KNN (Guo et al., 2003) is included as a reported classical machine-learning baseline based on instance-level similarity. It predicts a query drug-target pair from nearby training examples, with the reported comparison using $k = 5$. Unlike parametric neural models, KNN relies directly on the geometry of the feature space and therefore reflects how much local similarity among drug-target pairs is sufficient for prediction. This makes it a simple but informative reference for datasets where similar drugs or targets may recur across splits.

LR (Friedman et al., 2000) is included as a reported classical machine-learning baseline based on linear probabilistic classification. It maps weighted input features to an interaction probability through a sigmoid function. Because LR has limited capacity, its reported performance provides a lower-complexity reference for the discriminative strength of the input features alone. It also gives a calibrated probabilistic baseline for comparison with higher-capacity neural interaction models.

GraphDTA (Nguyen et al., 2021) is included as a reported graph-based neural baseline. It encodes molecular graphs with graph neural networks and protein sequences with convolutional layers before fully connected interaction prediction. This design represents a common DTI modeling pattern in which ligand topology and protein sequence motifs are encoded separately and fused only after feature extraction. It is therefore a relevant reference for evaluating whether more explicit interaction modeling improves over standard graph-plus-sequence encoding.

DrugBAN (Bai et al., 2023) is included as a reported graph- and attention-based DTI baseline. It uses bilinear attention to match drug and target representations and incorporates domain adaptation to improve transfer across data distributions. The bilinear attention module gives an explicit mechanism for pairwise drug-target matching, rather than relying only on concatenated representations. Its domain-adaptation component also makes it a useful reference for the cross-dataset distribution differences considered in this study.

IIFDTI (Cheng et al., 2022) is included as a reported attention-based DTI baseline. It separates interactive and independent features before predicting drug-target interactions. This separation is intended to distinguish pair-specific interaction evidence from features that mainly describe each molecule or protein independently. We include it as a reported reference for attention-based feature disentanglement in DTI prediction.

BINDTI (Peng et al., 2025) is included as a reported structure-aware attention baseline. It combines GCN-based drug graph encoding, ACmix-based protein encoding, and bidirectional multi-head interaction attention. Its bidirectional attention design explicitly models how drug-side and target-side representations attend to each other during prediction. This makes it one of the closest reported references for comparing against models that emphasize interaction-aware fusion.

TransformerCPI (Chen et al., 2020) is included as a reported Transformer-based DTI baseline. It applies sequence self-attention to compound-protein interaction prediction and evaluates the learned interaction patterns with label-reversal experiments. The model emphasizes attention over sequential representations, providing a contrast to graph-centered molecular encoders. It is included to cover DTI baselines where long-range dependency modeling is handled primarily through Transformer-style attention.

MolTrans (Huang et al., 2021) is included as a reported Transformer-based interaction baseline. It tokenizes drug and protein substructures and models their higher-order interactions with Transformer and convolutional modules. By operating on substructure-level tokens, MolTrans provides a different interaction granularity from whole-graph or whole-sequence encoders. This makes it a useful reported reference for evaluating whether explicit substructure interaction modeling is competitive with the proposed target-aware refinement strategy.

LDM-DTI (Ji et al., 2026) is included as the reported multimodal reference model. It combines pretrained drug and protein representations with graph topology, molecular geometry, protein local-context encoding, and bidirectional interaction fusion. We use it as a close reported reference for multimodal DTI encoding and interaction modeling, while avoiding any claim that it was reproduced in our experiments. Table 4 therefore serves as a reported ranking-metric landscape rather than a paired significance comparison: the baseline rows are literature values, whereas TAMR-DTI is evaluated under our repeated-run protocol.

Table 4: AUROC/AUPRC comparison on four benchmark datasets. Baseline values are adopted from reported results; TAMR-DTI values are from our five-seed runs and are shown as mean $\pm$ standard deviation.

	BindingDB		BioSNAP		Human		C. elegans	
Method	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
SVM	0.904 ± 0.002	0.865 ± 0.001	0.819 ± 0.045	0.839 ± 0.038	0.913 ± 0.012	0.905 ± 0.001	0.925 ± 0.009	0.907 ± 0.002
RF	0.942 ± 0.001	0.923 ± 0.001	0.857 ± 0.009	0.872 ± 0.006	0.939 ± 0.009	0.927 ± 0.001	0.935 ± 0.009	0.931 ± 0.001
KNN	0.895 ± 0.001	0.865 ± 0.002	0.842 ± 0.003	0.805 ± 0.003	0.915 ± 0.002	0.902 ± 0.002	0.912 ± 0.003	0.896 ± 0.002
LR	0.916 ± 0.003	0.884 ± 0.002	0.821 ± 0.004	0.796 ± 0.002	0.895 ± 0.003	0.862 ± 0.001	0.892 ± 0.002	0.883 ± 0.003
GraphDTA	0.944 ± 0.004	0.925 ± 0.006	0.871 ± 0.004	0.870 ± 0.006	0.965 ± 0.003	0.955 ± 0.003	0.959 ± 0.004	0.951 ± 0.002
DrugBAN	0.952 ± 0.001	0.936 ± 0.001	0.902 ± 0.001	0.905 ± 0.002	0.979 ± 0.001	0.969 ± 0.005	0.981 ± 0.002	0.978 ± 0.002
IIFDTI	0.924 ± 0.002	0.901 ± 0.004	0.895 ± 0.003	0.898 ± 0.005	0.977 ± 0.003	0.967 ± 0.027	0.984 ± 0.002	0.979 ± 0.002
TransformerCPI	0.895 ± 0.002	0.861 ± 0.003	0.843 ± 0.008	0.859 ± 0.007	0.956 ± 0.005	0.943 ± 0.002	0.975 ± 0.003	0.975 ± 0.002
MolTrans	0.935 ± 0.003	0.901 ± 0.004	0.879 ± 0.006	0.882 ± 0.006	0.979 ± 0.003	0.975 ± 0.002	0.982 ± 0.002	0.979 ± 0.002
BINDTI	0.956 ± 0.001	0.942 ± 0.002	0.895 ± 0.003	0.894 ± 0.003	0.980 ± 0.001	0.975 ± 0.004	0.985 ± 0.002	0.984 ± 0.002
LDM-DTI	0.960 ± 0.002	0.945 ± 0.002	0.908 ± 0.003	0.906 ± 0.003	0.981 ± 0.001	0.976 ± 0.002	0.988 ± 0.002	0.986 ± 0.002
TAMR-DTI	$\mathbf{0.964 \pm 0.001}$	$\mathbf{0.954 \pm 0.002}$	$\mathbf{0.930 \pm 0.002}$	$\mathbf{0.935 \pm 0.002}$	$\mathbf{0.982 \pm 0.002}$	$\mathbf{0.978 \pm 0.002}$	$\mathbf{0.992 \pm 0.001}$	$\mathbf{0.991 \pm 0.001}$

4.3 Benchmark Performance and Dataset-Level Behavior

The benchmark results demonstrate the overall effectiveness of TAMR-DTI across the four DTI benchmarks. The primary comparison is based on AUROC and AUPRC, because these threshold-independent metrics directly measure how well a model ranks interacting drug-target pairs ahead of non-interacting pairs. This ranking setting is especially important for DTI prediction, where candidate interactions are often prioritized before a downstream decision threshold is chosen.

The baseline landscape summarized in Table 4 shows that TAMR-DTI obtains the strongest reported AUROC/AUPRC values on all four benchmarks. On BindingDB, TAMR-DTI improves the reported AUROC/AUPRC envelope from 0.960/0.945 to 0.964/0.954. On BioSNAP, the gain is larger, reaching 0.930 AUROC and 0.935 AUPRC. On Human, TAMR-DTI reaches 0.982 ± 0.002 AUROC and 0.978 ± 0.002 AUPRC, exceeding the strongest reported ranking references at the displayed precision. On C. elegans, the model further improves the ranking scores to 0.992 AUROC and 0.991 AUPRC. These results indicate that the proposed target-aware conformer aggregation, ligand-conditioned protein encoding, and protein-side Mamba refinement provide consistent ranking benefits across datasets with different sizes, label ratios, and organism sources.

The dataset-level behavior also matches the design motivation of TAMR-DTI. BioSNAP is nearly balanced and contains enough repeated drug-target structure for pair-specific multimodal interaction modeling to be effective; accordingly, TAMR-DTI shows clear improvements across both ranking and fixed-threshold metrics. BindingDB tests scalability to broader chemical and protein coverage, where TAMR-DTI improves the ranking metrics while maintaining strong fixed-threshold accuracy. Human is the smallest benchmark in this study, yet the model still achieves the best reported AUROC/AUPRC values, showing that the architecture remains effective when training support is more limited. C. elegans provides a non-human organism benchmark, and the strong reported AUROC/AUPRC results show that TAMR-DTI remains competitive when trained and evaluated on this non-human interaction dataset.

The repeated-run table reports the five-seed TAMR-DTI summary across all four metrics in Table 5. The AUROC and AUPRC columns give the main ranking evidence, while Acc@0.5 and F1@0.5 provide a fixed-threshold view of the same runs.

Table 5: TAMR-DTI repeated-run summary across the four benchmark datasets. Values are mean $\pm$ standard deviation over seeds 42, 43, 44, 45, and 46.

Dataset	AUROC	AUPRC	Acc@0.5	F1@0.5
BindingDB	0.964 ± 0.001	0.954 ± 0.002	0.909 ± 0.003	0.891 ± 0.004
BioSNAP	0.930 ± 0.002	0.935 ± 0.002	0.857 ± 0.004	0.859 ± 0.004
Human	0.982 ± 0.002	0.978 ± 0.002	0.930 ± 0.009	0.924 ± 0.009
C. elegans	0.992 ± 0.001	0.991 ± 0.001	0.961 ± 0.004	0.961 ± 0.004

Figure 3 summarizes the same comparison as an all-method uncertainty view over AUROC, AUPRC, Acc@0.5, and F1@0.5. Intervals are computed from the available mean and standard deviation summaries using the same five-run t-interval formula when those summaries are available; for TAMR-DTI, intervals are computed from our five seeds. This view keeps the main ranking evidence and the fixed-threshold operating behavior in one visual summary, while Table 4 provides the exact AUROC/AUPRC values.

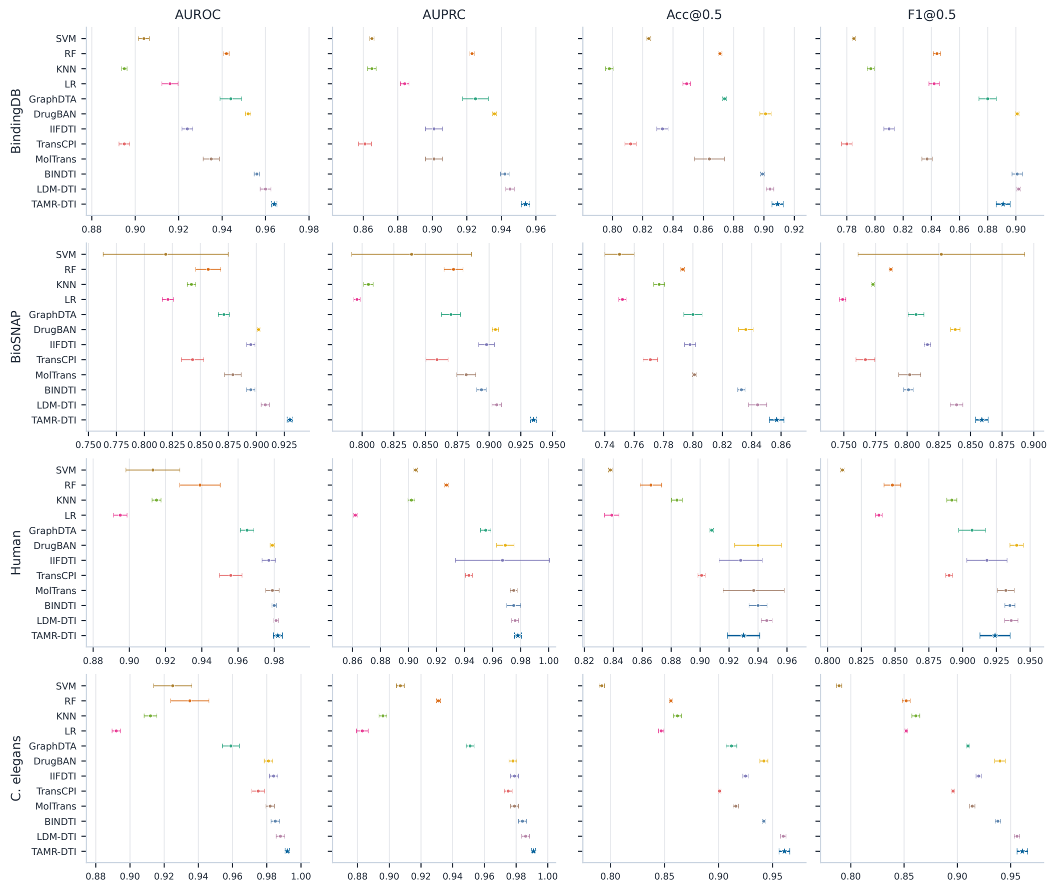

Figure 3: All-method uncertainty comparison on AUROC, AUPRC, Acc@0.5, and F1@0.5. Intervals are derived from mean and standard deviation summaries; TAMR-DTI intervals are computed from five local seeds.

Overall, the main benchmark results establish TAMR-DTI as a strong multimodal DTI predictor, with the clearest advantage in threshold-independent ranking performance. The fixed-threshold metrics complement this conclusion by showing the operating behavior at $\tau = 0.5$, while the AUROC/AUPRC results provide the central evidence for improved drug-target interaction ranking.

4.4 Stability Across Seeds

The repeated-seed analysis separates stable ranking gains from split-sensitive behavior. BindingDB, BioSNAP, and C. elegans form compact AUROC/AUPRC clusters, indicating that the ranking improvements are not driven by a single favorable seed. Human also stays above the reported-reference ranking range across most seeds, while its fixed-threshold metrics vary more strongly than the larger benchmarks. This suggests that the Human result should be interpreted together with calibration and seed-level variation, even though its mean AUROC and AUPRC are competitive with the strongest reported references.

The seed-level view in Figure 4 is used for this interpretation. The plot is descriptive rather than inferential: it shows where the proposed model is stable enough to support a benchmark claim and where additional calibration and repeated-run robustness work is still needed.

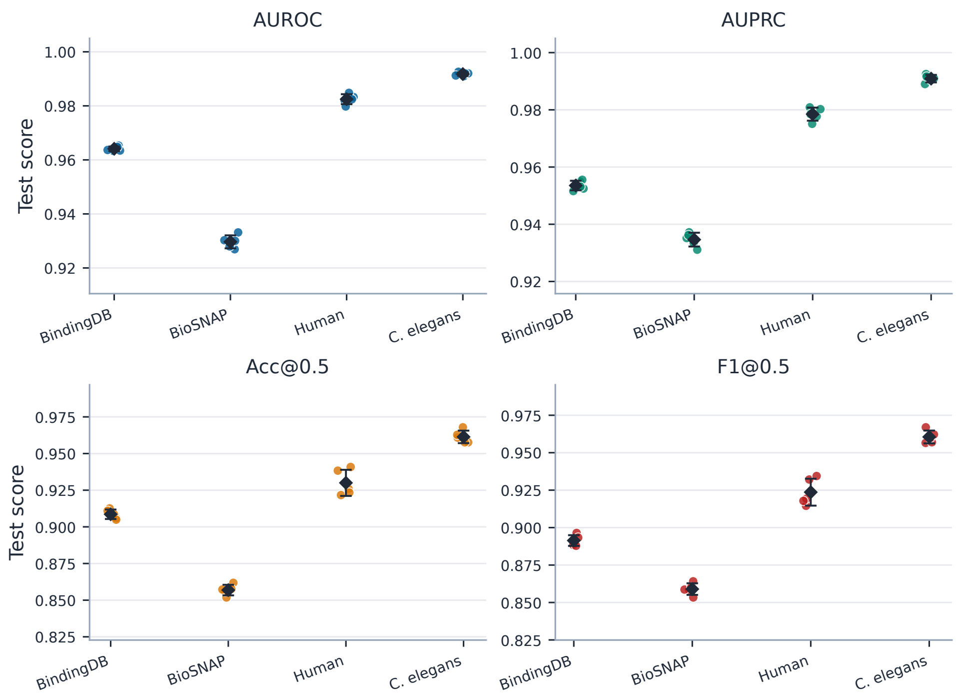

Figure 4: Five-seed stability of TAMR-DTI on the four benchmark datasets. Dots denote individual seeds 42, 43, 44, 45, and 46; diamonds denote seed means; and error bars denote one standard deviation. Acc@0.5 and F1@0.5 use the fixed threshold 0.5.

4.5 Ablation, Conformer Comparison, and Interpretability

The BioSNAP gain is analyzed through the mechanism that TAMR-DTI is designed to implement rather than through separate table-by-table comparisons. The initial protein representation first conditions the selection of drug conformers, so the small-molecule geometry branch is not averaged independently of the target. The aggregated drug context then feeds back into protein encoding through FiLM modulation, protein-side Mamba refines the longer sequence-token stream, and the final bidirectional interaction module aligns the two modalities before prediction. We examine this chain with module removal and conformer-count controls, and we restrict the interpretability analysis to the conformer-weight distribution produced by the target-aware aggregation module. BioSNAP is used because it is nearly balanced and gives the clearest ranking-metric improvement in the main benchmark comparison.

4.5.1 Module ablation

The module ablation asks whether the BioSNAP result comes from the intended cross-modal conditioning path or from a single oversized encoder. We conduct a single-seed diagnostic study with seed 42, the same train/validation/test split, the same validation-AUROC checkpoint-selection rule, and the same training configuration as the full model. Each ablated variant removes one module while keeping the remaining architecture unchanged. Because the ablation is not repeated over multiple seeds, we treat it as a diagnostic module-necessity analysis and report absolute scores rather than inferential claims. Acc@0.5 and F1@0.5 are computed at the fixed threshold 0.5.

The full model forms the upper envelope in Table 6, reaching 0.933 AUROC, 0.937 AUPRC, 0.862 Acc@0.5, and 0.864 F1@0.5. The largest AUROC loss appears when target-aware conformer aggregation is removed, dropping from 0.933 to 0.916 and reducing F1@0.5 from 0.864 to 0.845. This supports the core geometry argument: conformers should not be treated as target-agnostic views, because the relevant molecular geometry depends on the protein context. Removing ligand-conditioned protein FiLM gives the largest fixed-threshold accuracy drop, from 0.862 to 0.842, which is consistent with the feedback direction in the architecture: after protein-guided conformer aggregation, the resulting drug context should still be allowed to modulate protein encoding before final interaction prediction.

Table 6: Single-seed diagnostic ablation results on BioSNAP. All variants use seed 42, the same split, validation-AUROC checkpoint selection, and fixed-threshold Acc@0.5/F1@0.5.

Variant	AUROC	AUPRC	Acc@0.5	F1@0.5
Full TAMR-DTI	**0.933**	**0.937**	**0.862**	**0.864**
w/o Target-aware conformer	0.916	0.922	0.844	0.845
w/o Protein FiLM	0.920	0.924	0.842	0.844
w/o Protein Mamba refinement	0.918	0.925	0.842	0.845
w/o Bidirectional modulation	0.920	0.926	0.848	0.849

Reading the ablation scores as drops from the full model makes the component roles clearer. The relative-drop view in Figure 5 shows that removing the target-aware conformer module produces the largest AUROC drop, removing ligand-conditioned FiLM produces the largest Acc@0.5 drop, and removing protein Mamba refinement or bidirectional modulation yields consistent but slightly smaller degradation across the four metrics. The full model is included as the zero-drop reference, so the plot directly shows how far each ablated configuration moves from the complete TAMR-DTI setting.

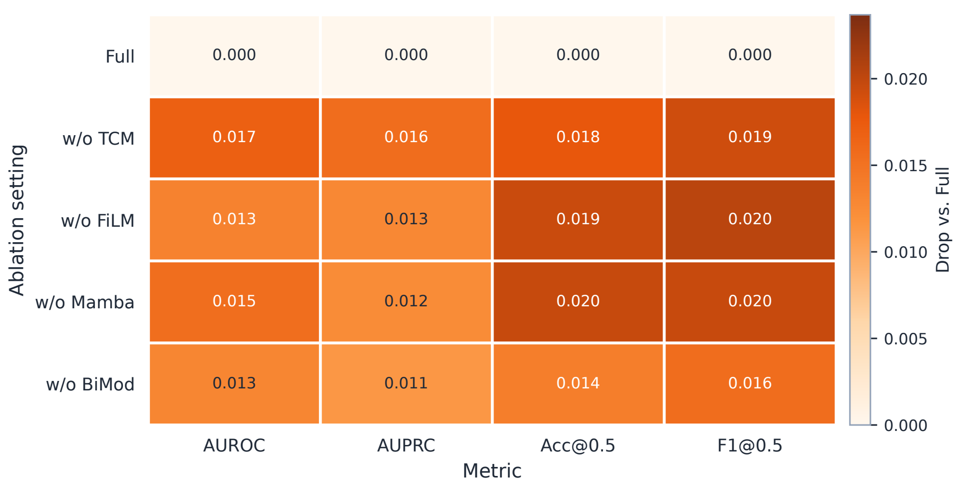

Figure 5: Single-seed diagnostic ablation drops on BioSNAP relative to the full TAMR-DTI model. Each cell reports the relative test-score decrease caused by removing one module under the same seed, split, validation-AUROC checkpoint selection, and fixed-threshold Acc@0.5/F1@0.5; the full model is shown as the zero-drop reference.

The protein Mamba and bidirectional-interaction ablations complete the same picture. Removing protein Mamba refinement lowers AUROC to 0.918 and F1@0.5 to 0.845, showing that the protein branch loses discriminative information when its token sequence is not further refined. This placement is intentional: TAMR-DTI applies Mamba to protein tokens rather than to SMILES tokens, because the drug branch is already represented through graph, pretrained, and conformer-level views, while the protein branch provides the longer sequence-token substrate where state-space refinement has a clearer role. Removing bidirectional modulation gives the smallest but still consistent degradation, with AUROC decreasing to 0.920 and F1@0.5 decreasing to 0.849. The four modules therefore act as complementary parts of one conditioning path: target-aware geometry selection, drug-to-protein FiLM feedback, protein-side sequential refinement, and final cross-modal alignment.

4.5.2 CONFORMER-COUNT COMPARISON

The conformer-count comparison isolates where the geometry branch helps. We compare six BioSNAP seed-42 variants that differ in the effective number of valid conformers or in the conformer aggregation rule. The $n = 1$, $n = 2$, and $n = 4$ variants are derived from the same eight-slot conformer cache by keeping only the first n valid conformer slots and masking the rest. The $n = 8$ variant uses the full eight-conformer cache used in the main model, and the $n = 16$ variant uses a separately generated sixteen-conformer cache to test whether a larger candidate geometry set improves the model further. Because preprocessing sorts generated conformers by computed energy, conformer count also controls the energy range exposed to the model: $n = 1$ keeps only the lowest-energy conformer, $n = 2$ and $n = 4$ add lower-energy alternatives, $n = 8$ exposes a broader geometry set, and $n = 16$ adds still more candidate conformers that may also contain additional irrelevant geometry. The $n = 8$ uniform-average variant keeps the same eight-conformer input but disables target-aware scoring and therefore serves as the aggregation control. All variants use the same data split, seed, optimizer, and validation-AUROC checkpoint selection; this experiment is interpreted as a conformer-count diagnostic rather than a hyperparameter search. The early-stopping patience is set to 100 epochs in this diagnostic comparison so that all variants can complete the full training horizon; Acc@0.5 and F1@0.5 are computed at the fixed threshold 0.5.

Table 7: Single-seed conformer-count comparison on BioSNAP. Conformers are energy-sorted during preprocessing; $n = 1$ keeps the lowest-energy conformer only, while larger n values expose additional geometries.

Variant	Effective conformers	Aggregation	Best epoch	AUROC	AUPRC	Acc@0.5	F1@0.5
$n = 1$ target-aware	1	Target-aware	46	0.9277	0.9297	0.8578	0.8591
$n = 2$ target-aware	2	Target-aware	68	0.9300	0.9322	0.8575	0.8605
$n = 4$ target-aware	4	Target-aware	46	0.9297	0.9345	0.8524	0.8542
$n = 8$ target-aware	8	Target-aware	51	**0.9322**	**0.9356**	0.8575	0.8624
$n = 16$ target-aware	16	Target-aware	62	0.9312	0.9326	**0.8631**	**0.8668**
$n = 8$ uniform average	8	Uniform average	64	0.9308	0.9340	0.8589	0.8616

The results in Table 7 show that the eight-conformer target-aware variant gives the strongest ranking performance, reaching 0.9322 AUROC and 0.9356 AUPRC. The single-conformer variant reaches 0.9277 AUROC and 0.9297 AUPRC, which suggests that fixing the drug geometry to the lowest-energy conformer discards useful alternative spatial views. Increasing the candidate set to $n = 16$ does not improve the ranking metrics over $n = 8$: AUROC decreases from 0.9322 to 0.9312 and AUPRC decreases from 0.9356 to 0.9326, although fixed-threshold Acc@0.5 and F1@0.5 increase in this single-seed run. This explains why the main experiments use $K = 8$: it provides the best AUROC/AUPRC tradeoff for the ranking-focused benchmark claim, while avoiding the additional geometry noise and compute cost introduced by sixteen conformers. The intermediate $n = 2$ and $n = 4$ variants also do not improve monotonically across all metrics, so the result should not be read as "more conformers always win." Instead, it supports a more specific interpretation: energy-sorted conformers provide a candidate geometry set, and target-aware aggregation is needed to decide which conformers are useful for a given protein.

The absolute-score view in Figure 6 highlights the same tradeoff. The target-aware $n = 8$ variant is highest on AUROC and AUPRC, whereas $n = 16$ is higher on the fixed-threshold Acc@0.5 and F1@0.5 in this diagnostic run. We therefore interpret the conformer-count comparison conservatively: multiple conformers are useful, and protein-conditioned weighting improves ranking metrics up to the eight-conformer setting, but the fixed-threshold metrics do not support a claim that a larger conformer set dominates at every operating point.

Together with the benchmark comparison, these diagnostics explain why BioSNAP is the strongest positive case for TAMR-DTI. The gain is not attributed to a single module in isolation. It depends on protein-guided conformer selection, drug-context feedback into protein encoding, protein-side sequential refinement, and final bidirectional alignment. The conformer-count result also places the energy issue in the right scope: the lowest-energy conformer is a useful starting point, but binding prediction benefits from exposing additional geometries and letting the target representation weight them.

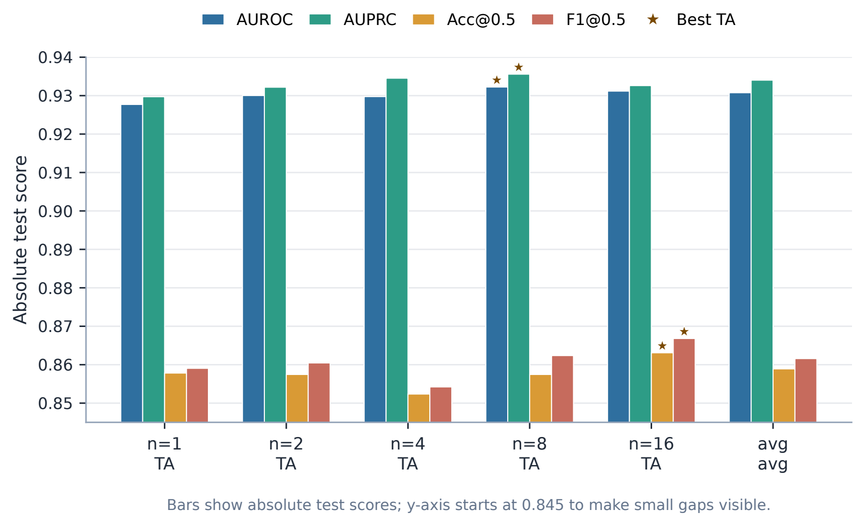

Figure 6: Absolute-score comparison of BioSNAP conformer-count variants, including the sixteen-conformer diagnostic run. TA denotes target-aware conformer aggregation, and avg denotes uniform averaging. The y-axis starts from 0.845 to make the small metric gaps visible; Acc@0.5 and F1@0.5 use the fixed threshold 0.5.

4.5.3 Interpretability analysis

The conformer-count comparison shows that exposing multiple candidate geometries is useful, but it does not reveal how the final BioSNAP model uses those geometries at inference time. We therefore perform a post-hoc forward-pass analysis with the fixed BioSNAP seed-42 checkpoint (best checkpoint at epoch 46, validation AUROC 0.9302, test AUROC 0.9321). No additional model or explainer is trained, and no parameter update is performed for this analysis. Instead, we run the held-out BioSNAP test pairs through the checkpoint once and collect the target-conditioned conformer weights emitted by the protein-conditioned aggregation module. The validation-selected threshold $\tau_{\text{val}} = 0.62$ is used only to assign TP/TN/FP/FN groups for stratified diagnostics. The resulting analysis covers all 5,493 BioSNAP test pairs.

For each drug-target pair, the forward pass assigns a probability distribution over the valid conformers in the precomputed conformer cache. Let $\mathbf{w}_i = \{w_{i1}, \ldots, w_{iK_i}\}$ denote these aggregation weights for test pair i, where K_i is the number of valid conformers. We quantify the sharpness of target-conditioned conformer usage by the normalized entropy

$$\bar{H}_i = -\frac{1}{\log K_i} \sum_{k=1}^{K_i} w_{ik} \log(w_{ik} + \epsilon), \tag{43}$$

where ϵ is a small constant for numerical stability. A smaller $\bar{H}_i$ indicates that the forward prediction concentrates on a small subset of conformers, whereas a larger value indicates softer multi-conformer aggregation.

To avoid cherry-picking individual examples, we summarize the within-sample sorted-weight distribution over the entire test set. Figure 7 reports, separately for true positives ($n = 2175$), true negatives ($n = 2328$), false positives ($n = 417$), and false negatives ($n = 573$), the median, IQR, and 5–95% band of the conformer-weight value at each rank after sorting the weights in descending order within each sample. Across all test pairs, the mean top-1 conformer weight is 0.597 and the mean top-1 margin is 0.365. The mean normalized conformer entropy is 0.480, showing that inference-time aggregation is clearly non-uniform while still retaining a graded contribution from multiple conformers.

The prediction-group statistics suggest one inference-time failure pattern. False negatives have the sharpest conformer distributions, with higher top-1 weight (0.631) and lower normalized entropy (0.445) than true positives (top-1 0.591, entropy 0.488) and true negatives (top-1 0.597, entropy

0.481). A coarse quartile comparison shows the same tendency without relying on the TP/TN/FP/FN grouping: samples in the lowest entropy quartile ($\bar{H} \leq 0.378$) have a higher error rate than samples in the highest entropy quartile ($\bar{H} \geq 0.648$), 19.6% versus 16.3%. This observation should be read as diagnostic rather than causal: the conformer branch is most useful not when it collapses to a single dominant geometry, but when it uses the protein context to retain an appropriate mixture of plausible conformers.

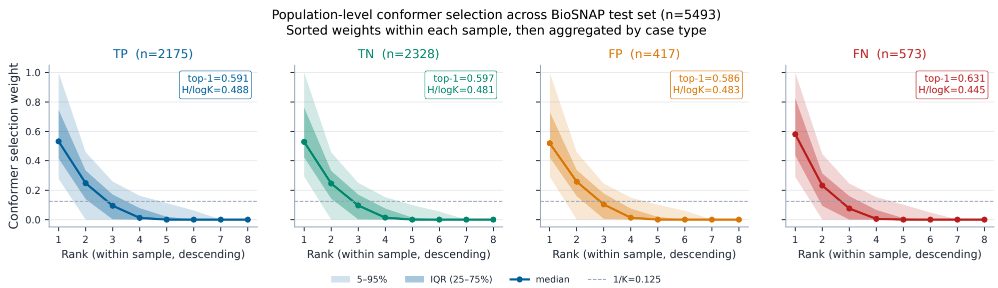

Figure 7: Population-level target-aware conformer-weight distribution across the entire BioSNAP test set ($n = 5,493$). For each test pair, the $K = 8$ conformer weights are sorted in descending order within the sample, and the resulting rank-vs-weight curves are aggregated by prediction group (TP, TN, FP, FN). Solid curves and points are medians, dark bands are interquartile ranges, and light bands are the 5–95% range; the dashed reference at $1/K = 0.125$ marks the uniform-weight baseline. Each panel also reports the per-group mean top-1 weight and mean normalized entropy.

5 DISCUSSION

The main results suggest that TAMR-DTI is most useful when the prediction task benefits from pair-specific representation learning. Instead of treating the drug and protein as two fixed encodings that are fused only at the end, the model lets the protein context influence conformer aggregation and lets the drug context condition protein encoding before final interaction prediction. This design matches the structure of DTI prediction: the same ligand can expose different useful geometric views for different targets, and the same protein sequence can carry different interaction evidence depending on the ligand. The benchmark results therefore support the value of target-aware modulation, while the threshold-dependent metrics remind us that ranking quality and calibrated decision behavior are related but distinct goals.

The conformer-count analysis should be read in this light. It does not imply that adding more conformers will always improve performance. Conformers are computationally generated geometric hypotheses, and later conformer slots may include less relevant or noisier geometries. The useful point is narrower: binding prediction can benefit from exposing multiple plausible molecular geometries when the model also has a target-conditioned mechanism for weighting them. This is why the conformer branch is better viewed as a protein-guided geometry aggregation module than as a simple multi-conformer augmentation trick.

The protein-side refinement modules play a complementary role. FiLM modulation injects ligand information into protein feature extraction, while the Mamba block refines the ordered protein-token stream before cross-modal matching. TAMR-DTI deliberately keeps attention-based interaction modeling in the final fusion stage, because DTI prediction still requires explicit drug-protein alignment. Mamba is therefore used as a controlled residual refinement on the protein side, not as a wholesale replacement for cross-modal attention. The ablation results are the main evidence for this design choice. The retained interpretability analysis complements this picture from the geometry side by showing that conformer usage is sample-specific across all 5,493 BioSNAP test pairs, with the lowest-entropy conformer selections showing a higher error rate than the highest-entropy selections. These visualizations should still be read as model-internal evidence rather than as direct structural validation.

Several limitations remain. The reported baselines come from the literature and do not provide a fully matched paired-testing protocol. The ablation and conformer-count studies are single-seed diagnostics on BioSNAP, so they support mechanistic interpretation but should not be treated as formal significance tests. The generated conformers are not experimentally resolved binding poses, and the current random-split benchmarks do not fully test scaffold-level or target-family generalization. Consequently, these experiments evaluate within-dataset prediction rather than cold-drug, cold-target, scaffold-level, or cross-organism transfer. The interpretability analysis is also limited to model-internal conformer-weight behavior and does not validate the generated conformers as experimental binding poses.

Future work should therefore use matched multi-seed ablations, harder out-of-distribution splits, calibration-aware threshold selection, and stronger structural supervision when reliable binding-pose information is available. Another practical direction is to improve the generation of multi-conformer features themselves. Rather than relying only on a fixed energy-sorted conformer cache, future variants could use more diverse conformer sampling, better energy- or uncertainty-aware conformer scoring, and target-conditioned preselection or refinement of candidate geometries before aggregation. Such improvements would keep the target-aware weighting idea intact while reducing the noise introduced by irrelevant or low-quality generated conformers.

6 CONCLUSION

This work presented TAMR-DTI, a multimodal DTI framework that represents drug-target pairs through target-aware conformer modulation, ligand-conditioned protein encoding, protein-side Mamba refinement, and BiIntention-based interaction matching. Across five repeated runs, the model achieves the strongest reported AUROC/AUPRC values on all four benchmark datasets considered in this study: 0.964/0.954 on BindingDB, 0.930/0.935 on BioSNAP, 0.982/0.978 on Human, and 0.992/0.991 on C. elegans. These results show that the proposed representation path improves the primary ranking objective for DTI prediction across datasets with different sizes, label ratios, and organism sources.

The diagnostic results further clarify where the gains come from. On BioSNAP, removing target-aware conformer aggregation lowers AUROC from 0.933 to 0.916, while removing ligand-conditioned protein FiLM or protein-side Mamba refinement also reduces both ranking and fixed-threshold performance. The conformer- count comparison shows that the eight-conformer target-aware variant reaches the best ranking scores among the tested geometry settings (0.9322 AUROC and 0.9356 AUPRC), and the conformer-weight analysis shows non-uniform, sample-specific conformer usage over the held-out test pairs. Together, these findings support the central design conclusion: ligand geometry should be treated as pair-level evidence conditioned on the protein context, while long-range protein refinement should complement rather than replace explicit drug- target matching.

The broader significance is that DTI models can move beyond independently encoded drug and protein descriptors toward pair-specific representation construction. More concretely, TAMR-DTI provides a practical way to select and aggregate generated ligand conformers according to the target, feed the resulting ligand context back into protein representation learning, refine the ordered protein sequence with an efficient state-space block, and retain an explicit bidirectional interaction module for prediction. This gives future DTI systems a modular template for combining molecular geometry, protein sequence context, and interpretable pairwise matching. In practical drug-discovery workflows, such a model can help prioritize compound-target pairs for downstream validation, narrow the candidate space before costly biochemical assays, and provide model-internal conformer-usage evidence for why a ligand may be more plausible for one target than another, while still requiring experimental validation before biological or clinical use. The current evidence is strongest for ranking performance; future work should strengthen the conclusion with matched multi-seed ablations, harder generalization splits, and more direct structural supervision for conformer selection.

Supporting Information

Additional methodological details; standalone high-resolution versions of the figures and tables; equation and notation guide for the TAMR-DTI architecture (PDF).

Data Availability Statement

Code: `https://github.com/leleleocc/TAMR-DTI`. Datasets: BindingDB (`https://www.bindingdb.org/`), BioSNAP (`https://snap.stanford.edu/biodata/`), and Human/C. elegans data (`https://github.com/lifanchen-simm/transformerCPI`).

Acknowledgments

This work was supported by the Ministry of Science and Technology of the People's Republic of China (No. 2025YFC3408501).

References

Stephen F. Altschul, Thomas L. Madden, Alejandro A. Schäffer, Jinghui Zhang, Zheng Zhang, Webb Miller, and David J. Lipman. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25(17):3389–3402, 1997. doi: 10.1093/nar/25.17.3389.

Michael Ashburner, Catherine A. Ball, Judith A. Blake, David Botstein, Heather Butler, J. Michael Cherry, Allan P. Davis, Kara Dolinski, Selina S. Dwight, Janan T. Eppig, et al. Gene ontology: tool for the unification of biology. *Nature Genetics*, 25(1):25–29, 2000. doi: 10.1038/75556.

Peizhen Bai, Filip Miljkovic, Brian John, and Hui Lu. Interpretable bilinear attention network with domain adaptation improves drug-target prediction. *Nature Machine Intelligence*, 5(2):126–136, 2023. doi: 10.1038/s42256-022-00605-1.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Lifan Chen, Xiaoqin Tan, Dan Wang, Feisheng Zhong, Xiaohong Liu, Tianbiao Yang, Xuan Luo, Kaixian Chen, Hualiang Jiang, and Mingyue Zheng. Transformercpi: Improving compound-protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments. *Bioinformatics*, 36(16):4406–4414, 2020. doi: 10.1093/bioinformatics/btaa524.

Zhongjian Cheng, Qichang Zhao, Yanjie Li, Jihua Wang, and Karsten Borgwardt. Iifdti: Predicting drug-target interactions through interactive and independent features based on attention mechanism. *Bioinformatics*, 38(17):4153–4161, 2022. doi: 10.1093/bioinformatics/btac485.

Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. Chemberta: Large-scale self-supervised pretraining for molecular property prediction. *arXiv preprint arXiv:2010.09885*, 2020.

Kuo-Chen Chou. Prediction of protein cellular attributes using pseudo-amino acid composition. *Proteins: Structure, Function, and Genetics*, 43(3):246–255, 2001. doi: 10.1002/prot.1035.

Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995.

Inna Dubchak, Ilya Muchnik, Stephen R. Holbrook, and Sung-Hou Kim. Prediction of protein folding class using global description of amino acid sequence. *Proceedings of the National Academy of Sciences*, 92(19):8700–8704, 1995. doi: 10.1073/pnas.92.19.8700.

David K. Duvenaud, Dougal Maclaurin, Jorge Aguilera-Iparraguirre, Rafael Gómez-Bombarelli, Timothy Hirzel, Alán Aspuru-Guzik, and Ryan P. Adams. Convolutional networks on graphs for learning molecular fingerprints. In *Advances in Neural Information Processing Systems*, volume 28, pp. 2224–2232, 2015.

Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. Prottrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381.

Robert D. Finn, Alex Bateman, Jody Clements, Penelope Coggill, Ruth Y. Eberhardt, Sean R. Eddy, Andreas Heger, Kirstie Hetherington, Liisa Holm, Jaina Mistry, Erik L. L. Sonnhammer, John Tate, and Marco Punta. Pfam: the protein families database. *Nucleic Acids Research*, 42(D1): D222–D230, 2014. doi: 10.1093/nar/gkt1223.

Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Additive logistic regression: A statistical view of boosting. *The Annals of Statistics*, 28(2):337–407, 2000.

Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pp. 1263–1272, 2017.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

Albert Gu, Karan Goel, and Christopher Re. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.

Gongde Guo, Hui Wang, David Bell, Yaxin Bi, and Kieran Greer. Knn model-based approach in classification. In *On The Move to Meaningful Internet Systems 2003: CoopIS, DOA, and ODBASE*, pp. 986–996, 2003.

Tong He, Marten Heidemeyer, Fang Ban, Artem Cherkasov, and Martin Ester. Simboost: a read-across approach for predicting drug-target binding affinities using gradient boosting machines. *Journal of Cheminformatics*, 9:24, 2017. doi: 10.1186/s13321-017-0209-z.

Yang Hua, Zhenhua Feng, Xiaoning Song, Xiao-Jun Wu, and Josef Kittler. Mmdg-dti: Drug-target interaction prediction via multimodal feature fusion and domain generalization. *Pattern Recognition*, 157:110887, 2025. doi: 10.1016/j.patcog.2024.110887.

Kexin Huang, Cao Xiao, Lucas M. Glass, and Jimeng Sun. Moltrans: Molecular interaction transformer for drug-target interaction prediction. *Bioinformatics*, 37(6):830–836, 2021. doi: 10.1093/bioinformatics/btaa880.

Yuanyuan Ji, Zhuo Chen, Zhihan Liu, Xiaofeng Man, Junwei Du, and Bin Yu. Ldm-dti: A multimodal framework integrating pretrained language models and geometric graph networks for interpretable drug-target interaction prediction. *Expert Systems with Applications*, 313:131485, 2026. doi: 10.1016/j.eswa.2025.131485.

Mingjian Jiang, Zhen Li, Shugang Zhang, Shuang Wang, Xiaofeng Wang, Qing Yuan, and Zhiqiang Wei. Drug-target affinity prediction using graph neural network and contact maps. *RSC Advances*, 10(35):20701–20712, 2020. doi: 10.1039/D0RA02297G.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.

Mostafa Karimi, Di Wu, Zhangyang Wang, and Yang Shen. Deepaffinity: interpretable deep learning of compound-protein affinity through unified recurrent and convolutional neural networks. *Bioinformatics*, 35(18):3329–3338, 2019. doi: 10.1093/bioinformatics/btz111.

Michael J. Keiser, Bryan L. Roth, Blaine N. Armbruster, Paul Ernsberger, John J. Irwin, and Brian K. Shoichet. Relating protein pharmacology by ligand chemistry. *Nature Biotechnology*, 25(2):197–206, 2007. doi: 10.1038/nbt1284.

Huan Yee Koh, Anh T. N. Nguyen, Shirui Pan, Lauren T. May, and Geoffrey I. Webb. PSICHIC: Physicochemical graph neural network for learning protein-ligand interaction fingerprints from sequence data. *Nature Machine Intelligence*, 6:728–739, 2024. doi: 10.1038/s42256-024-00847-1.

Ingoo Lee, Jongsoo Keum, and Hojung Nam. Deepconv-dti: prediction of drug-target interactions via deep learning with convolution on protein sequences. *PLoS Computational Biology*, 15(6):e1007129, 2019. doi: 10.1371/journal.pcbi.1007129.

Jonghyun Lee, Dae Won Jun, Ildae Song, and Yun Kim. Dlm-dti: a dual language model for the prediction of drug-target interaction with hint-based learning. *Journal of Cheminformatics*, 16:14, 2024. doi: 10.1186/s13321-024-00808-1.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

Hui Liu, Jianjiang Sun, Jihong Guan, Jie Zheng, and Shuigeng Zhou. Improving compound-protein interaction prediction by building up highly credible negative samples. *Bioinformatics*, 31(12):i221–i229, 2015. doi: 10.1093/bioinformatics/btv256.

Tiqing Liu, Yuhmei Lin, Xin Wen, Robert N. Jorissen, and Michael K. Gilson. Bindingdb: a web-accessible database of experimentally determined protein-ligand binding affinities. *Nucleic Acids Research*, 35(Suppl. 1):D198–D201, 2007. doi: 10.1093/nar/gkl999.

Yong Liu, Min Wu, Chunyan Miao, Peilin Zhao, and Xiao-Li Li. Neighborhood regularized logistic matrix factorization for drug-target interaction prediction. *PLoS Computational Biology*, 12(2):e1004760, 2016. doi: 10.1371/journal.pcbi.1004760.

Youzhi Liu, Linlin Xing, Longbo Zhang, Hongzhen Cai, and Maozu Guo. Geformerdta: drug target affinity prediction based on transformer graph for early fusion. *Scientific Reports*, 14:7416, 2024. doi: 10.1038/s41598-024-57879-1.

Zhangli Lu, Guoqiang Song, Huimin Zhu, Chuqi Lei, Xinliang Sun, Kaili Wang, Libo Qin, Yafei Chen, Jing Tang, and Min Li. Dtiam: a unified framework for predicting drug-target interactions, binding affinities and drug mechanisms. *Nature Communications*, 16:2548, 2025. doi: 10.1038/s41467-025-57828-0.

Yunan Luo, Xinbin Zhao, Jingtian Zhou, Jinglin Yang, Yanqing Zhang, Wenhua Kuang, Jian Peng, Ligong Chen, and Jianyang Zeng. A network integration approach for drug-target interaction prediction and computational drug repositioning from heterogeneous information. *Nature Communications*, 8:573, 2017. doi: 10.1038/s41467-017-00680-8.

Zhengchao Luo, Wei Wu, Qichen Sun, and Jinzhuo Wang. Accurate and transferable drug-target interaction prediction with druglamp. *Bioinformatics*, 40(12):btae693, 2024. doi: 10.1093/bioinformatics/btae693.

Thin Nguyen, Hang Le, Thomas P. Quinn, Tri Nguyen, Thuc Duy Le, and Svetha Venkatesh. Graphdta: Predicting drug-target binding affinity with graph neural networks. *Bioinformatics*, 37(8):1140–1147, 2021. doi: 10.1093/bioinformatics/btaa921.

Lihong Peng, Xiaoyan Liu, Long Yang, Longlong Liu, Zongzheng Bai, Xu Lu, and Libo Nie. Bindti: A bi-directional intention network for drug-target interaction identification based on attention mechanisms. *IEEE Journal of Biomedical and Health Informatics*, 29(3):1602–1612, 2025. doi: 10.1109/JBHI.2024.3375025.

Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

Ying Qian, Xinyi Li, Jian Wu, and Qian Zhang. Mcl-dti: using drug multimodal information and bi-directional cross-attention learning method for predicting drug-target interaction. *BMC Bioinformatics*, 24:323, 2023. doi: 10.1186/s12859-023-05447-1.

Na Quan, Shicheng Ma, Kai Zhao, Xuehua Bi, and Linlin Zhang. Mfcadti: improving drug-target interaction prediction by integrating multiple feature through cross attention mechanism. *BMC Bioinformatics*, 26:57, 2025. doi: 10.1186/s12859-025-06075-7.

Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021. doi: 10.1073/pnas.2016239118.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Victor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 9323–9332, 2021.

Maximilian G. Schuh, Davide Boldini, Annkathrin I. Bohne, and Stephan A. Sieber. BarlowDTI: advanced 1d drug-target interaction prediction. *Journal of Cheminformatics*, 17:18, 2025. doi: 10.1186/s13321-025-00952-2.

Kristof T. Schütt, Pieter-Jan Kindermans, Huziel Enoc Sauceda Felix, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Youyuan Shui, Xuewen Ge, Chen Cao, et al. Bima-dti: a bidirectional mamba-attention hybrid framework for enhanced drug-target interaction prediction. *BMC Biology*, 23:309, 2025. doi: 10.1186/s12915-025-02407-4.

Rohit Singh, Samuel Sledzieski, Bryan Bryson, Lenore Cowen, and Bonnie Berger. Contrastive learning in protein language space predicts interactions between drugs and protein targets. *Proceedings of the National Academy of Sciences*, 120(24):e2220778120, 2023. doi: 10.1073/pnas.2220778120.

Bosheng Song, Jiayi Zhang, Ying Liu, Yuansheng Liu, Jing Jiang, Sisi Yuan, Xia Zhen, and Yiping Liu. A systematic review of molecular representation learning foundation models. *Briefings in Bioinformatics*, 27(1):bbaf703, 2026. doi: 10.1093/bib/bbaf703.

Roberto Todeschini and Viviana Consonni. *Molecular Descriptors for Chemoinformatics*. Wiley-VCH, Weinheim, 2009.

Fangping Wan, Lixiang Hong, An Xiao, Tao Jiang, and Jianyang Zeng. Neodti: neural integration of neighbor information from a heterogeneous network for discovering new drug-target interactions. *Bioinformatics*, 35(1):104–111, 2019. doi: 10.1093/bioinformatics/bty543.

David Weininger. SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, 28(1):31–36, 1988. doi: 10.1021/ci00057a005.

Jun Xia, Chengshuai Zhao, Bozhen Hu, Zhangyang Gao, Cheng Tan, Yue Liu, Siyuan Li, and Stan Z. Li. Mole-bert: Rethinking pre-training graph neural networks for molecules. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=jevY-DtiZTR.

Yoshihiro Yamanishi, Michihiro Araki, Alex Gutteridge, Wataru Honda, and Minoru Kanehisa. Prediction of drug-target interaction networks from the integration of chemical and genomic spaces. *Bioinformatics*, 24(13):i232–i240, 2008. doi: 10.1093/bioinformatics/btn162.

Ziduo Yang, Weihe Zhong, Lu Zhao, and Calvin Yu-Chian Chen. Mgraphdta: deep multiscale graph neural network for explainable drug-target binding affinity prediction. *Chemical Science*, 13(3): 816–833, 2022. doi: 10.1039/D1SC05180F.

Qing Ye, Chang-Yu Hsieh, Ziyi Yang, Yu Kang, Jiming Chen, Dongsheng Cao, Shibo He, et al. A unified drug-target interaction prediction framework based on knowledge graph and recommendation system. *Nature Communications*, 12:6775, 2021. doi: 10.1038/s41467-021-27137-3.

Gwang-Hyeon Yun, Jong-Hoon Park, and Young-Rae Cho. Tridti: tri-modal representation learning with cross-modal alignment for drug-target interaction prediction. *Briefings in Bioinformatics*, 27 (1):bbag034, 2026. doi: 10.1093/bib/bbag034.

Yanpeng Zhao, Yuting Xing, Yixin Zhang, et al. Evidential deep learning-based drug-target interaction prediction. *Nature Communications*, 16:6915, 2025. doi: 10.1038/s41467-025-62235-6.

Gengmo Zhou, Zhifeng Gao, Qiankun Ding, Hang Zheng, Hongteng Xu, Zhewei Wei, Linfeng Zhang, and Guolin Ke. Uni-mol: A universal 3d molecular representation learning framework. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=6K2RM6wVqKu.

Marinka Zitnik, Rok Sosič, Sagar Maheshwari, and Jure Leskovec. BioSNAP Datasets: Stanford biomedical network dataset collection. http://snap.stanford.edu/biodata, aug 2018.